

Course Announcements

Due Sunday (11:59 PM):

- D7
- Q8
- Weekly Project Survey (*optional*)

Note: A4 will be released tomorrow (Tu 2/25)

Machine Learning

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Did they summarize the data? **Yes** → Did they report the summaries without interpretation? **No** → Did they quantify whether the discoveries are likely to hold in a new sample? **Yes** → Are they trying to figure out how changing the average of one measurement affects another?

Predictive: apply machine learning techniques to data you have currently to generate a model that will be able to make a prediction on future data

Classic Statistics
(parametric & nonparametric)

Text Analysis

Are they trying to predict measurement(s) for individuals?

Causal

STOP!
Not a data analysis

No Are the data a corpus of text?

No Are the observations spatially related?

Inferential

Predictive

Supervised Machine Learning

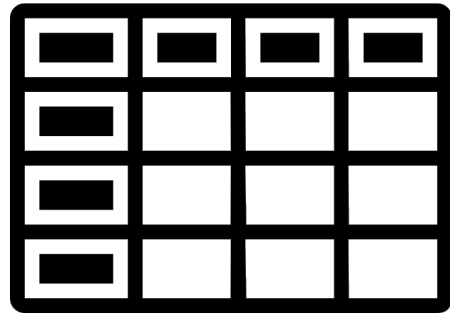
Unsupervised Machine Learning

Did the computer decide the features of your model?

Geospatial Statistics

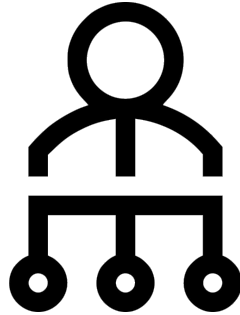
predictive analysis uses data
you have now to make
predictions in the future

machine learning approaches
are used for predictive analysis!



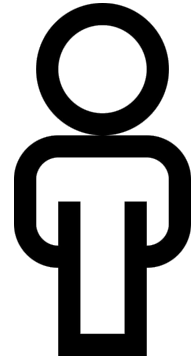
data

train →



model

→ predict



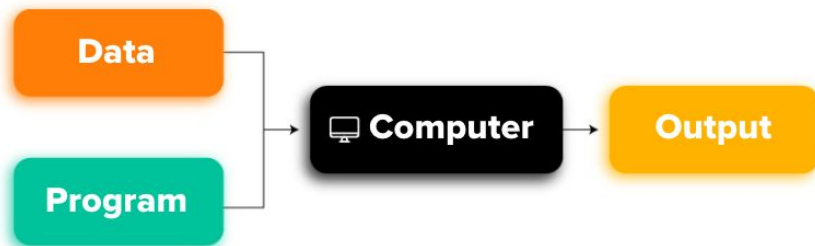
What is machine learning?

“Machine learning is the science of getting computers to act without being explicitly programmed”

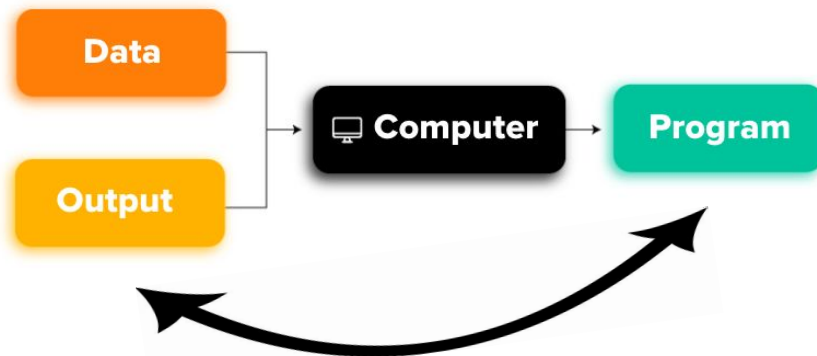
- Andrew Ng, Stanford, ex-Google, chief scientist at Baidu, Coursera founder, Stanford Adjunct Faculty

A Shift in Programming Paradigm

TRADITIONAL PROGRAMMING

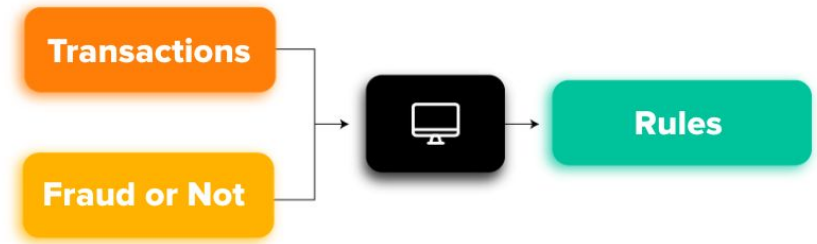
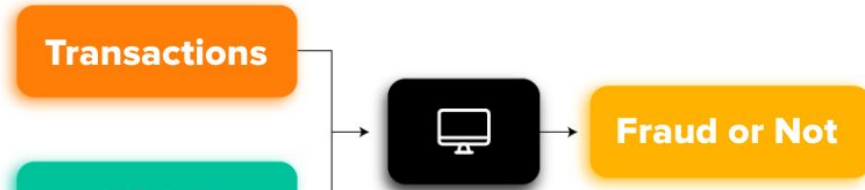


MACHINE LEARNING

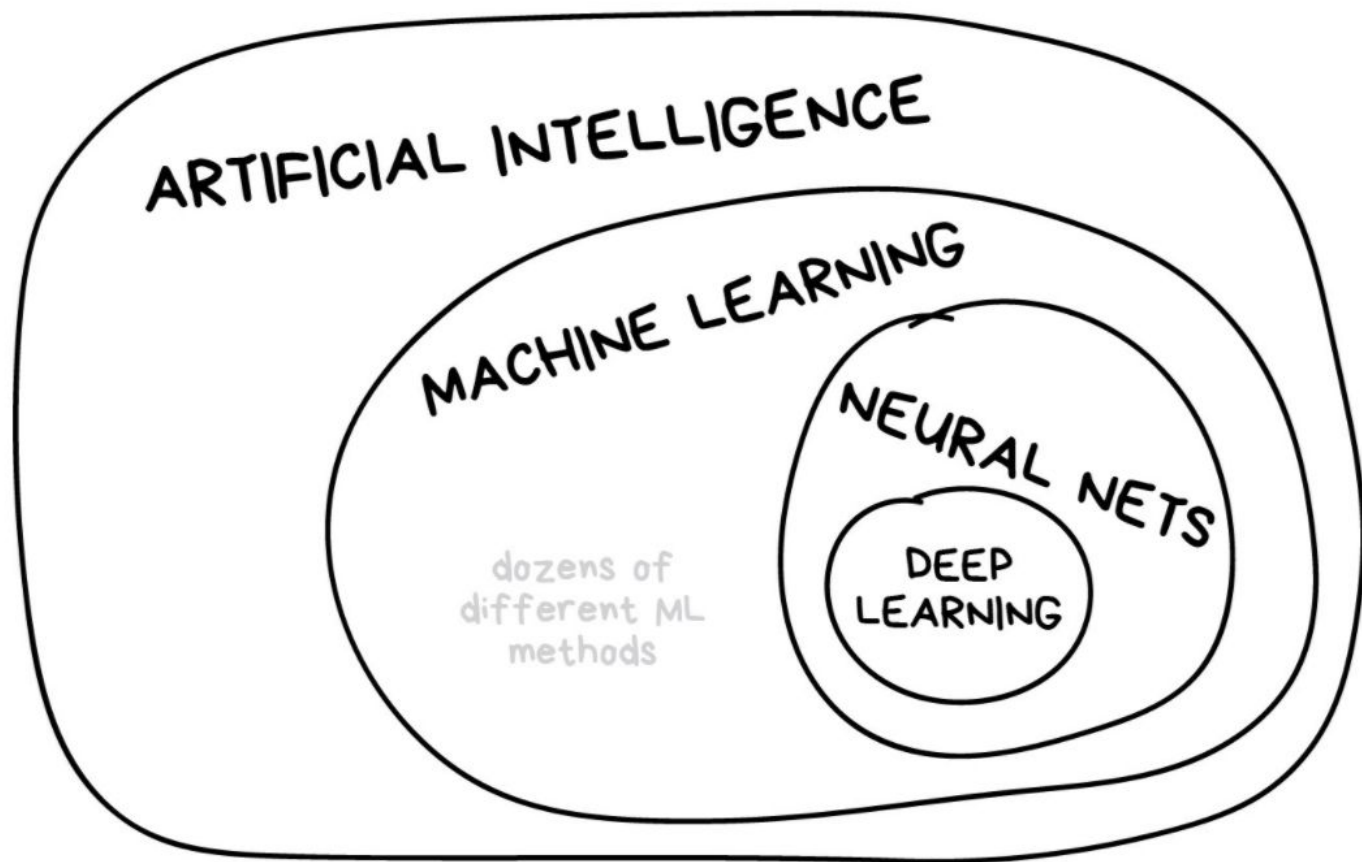


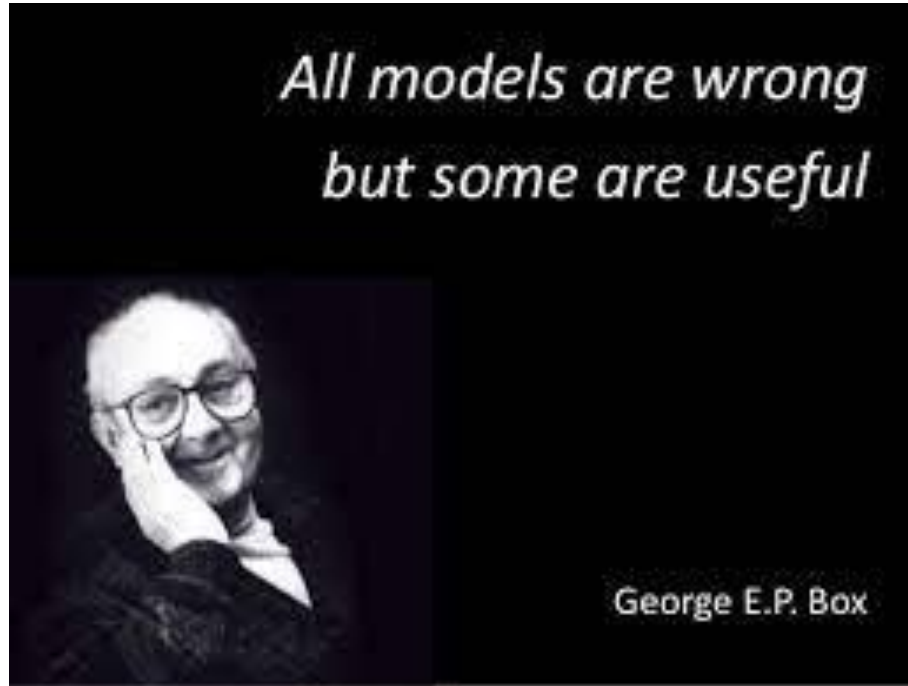
- **Problem:** Detecting whether credit card charges are fraudulent.
- **Data science question:** Can we use the time of the charge, the location of the charge, and the price of the charge to predict whether that charge is fraudulent or not?
- **Type of analysis:** Predictive analysis





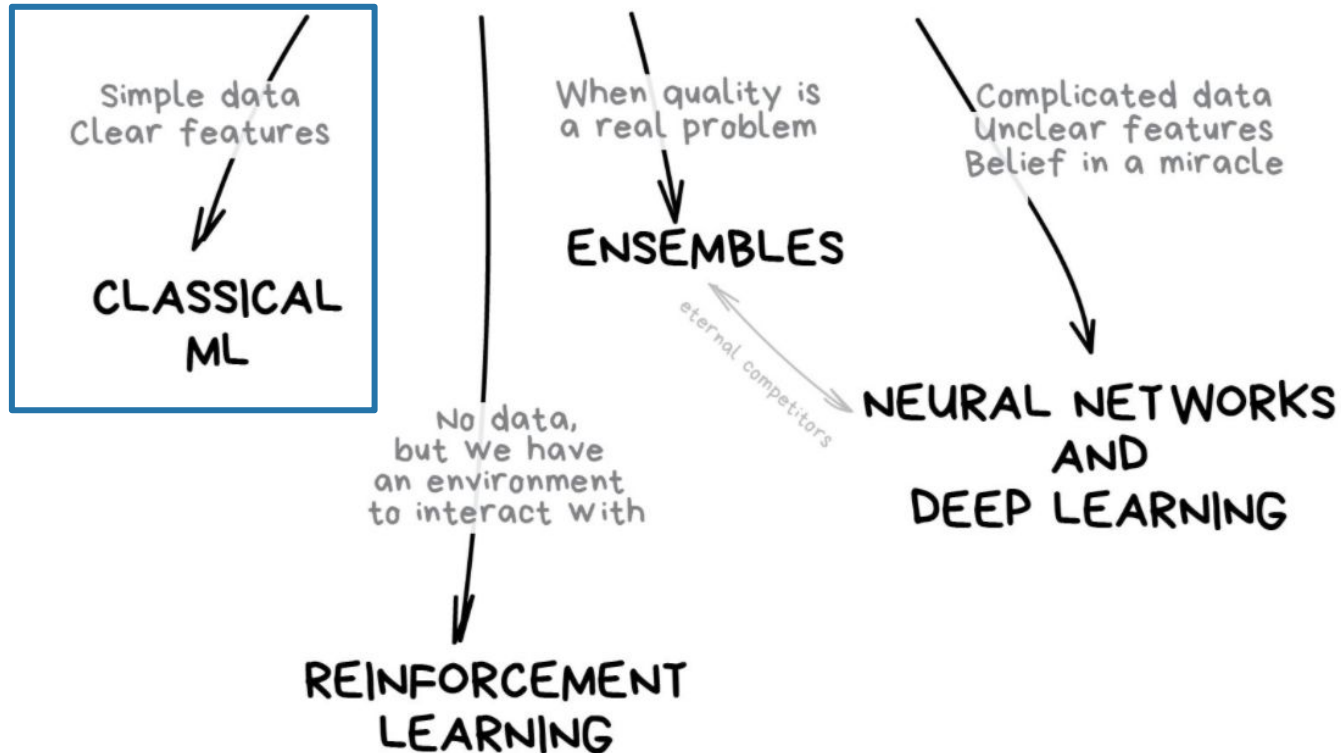
Rule1. Claim time - Submit time < 1 h
Rule2. Agreement review time > 5 m
Rule3. ...



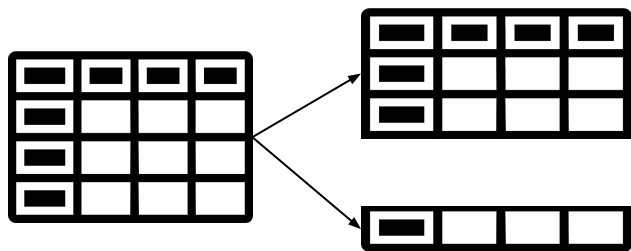


The goal of modeling is to simplify, not replicate

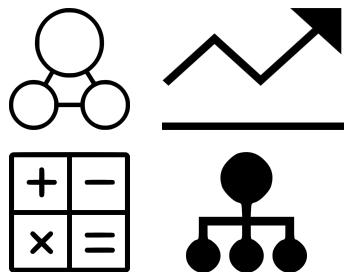
THE MAIN TYPES OF MACHINE LEARNING



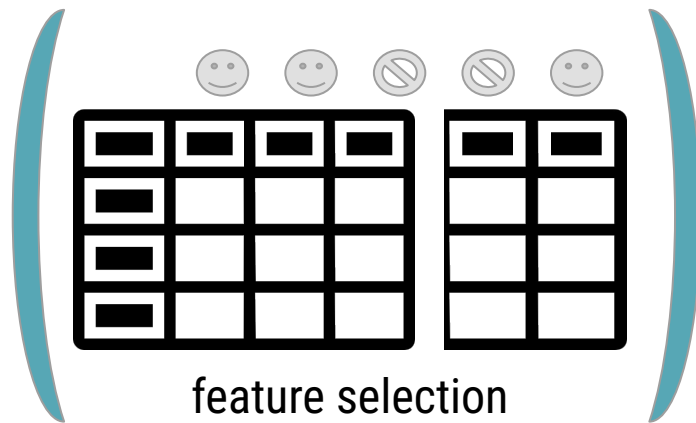
Basic Steps to Prediction



data
partitioning



model selection



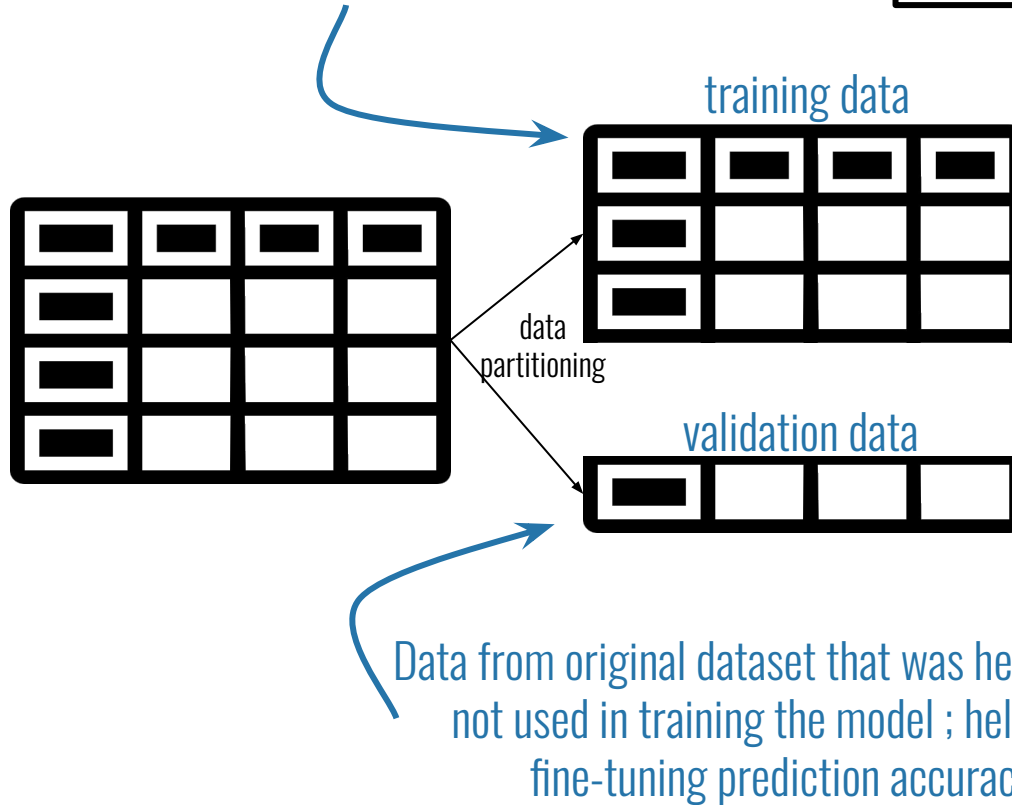
feature selection



model assessment

Data partitioning: Evaluating how you will do when you have huge amounts of data

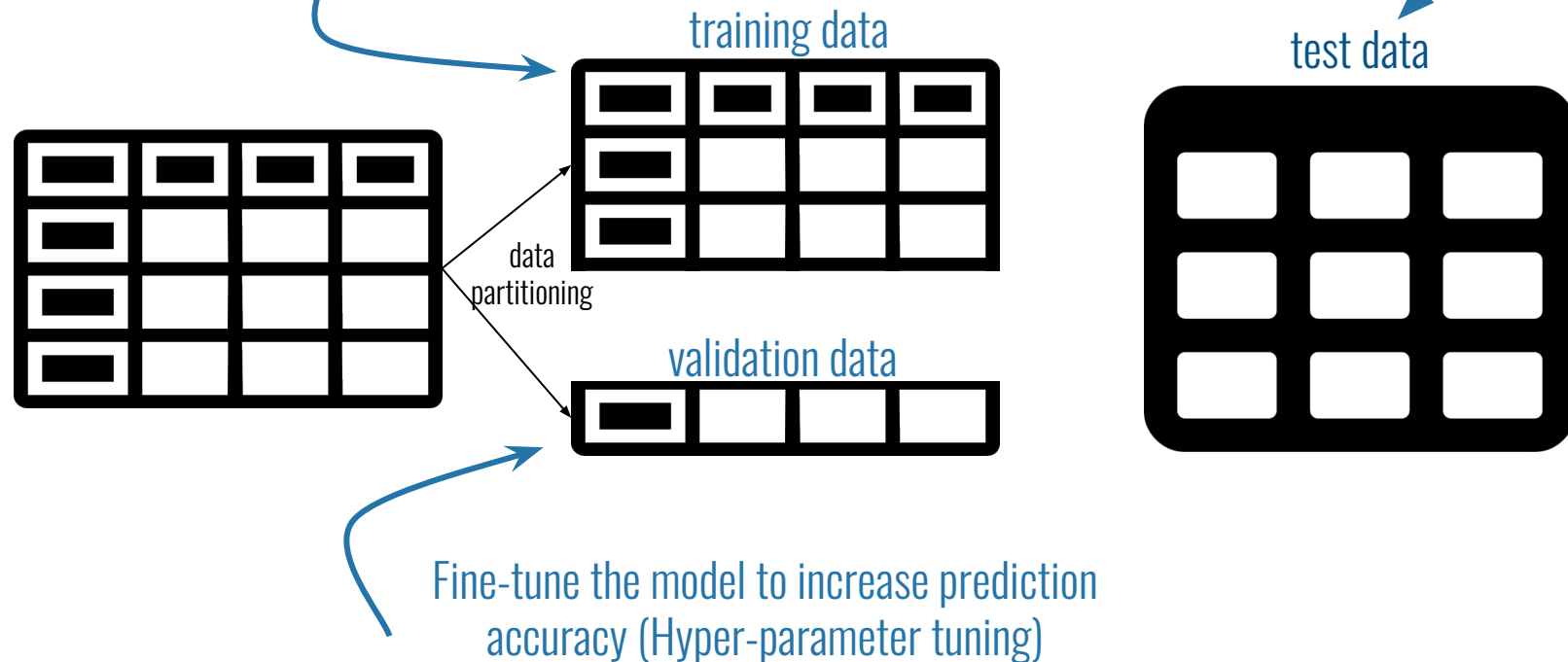
the data used to build
your predictive model

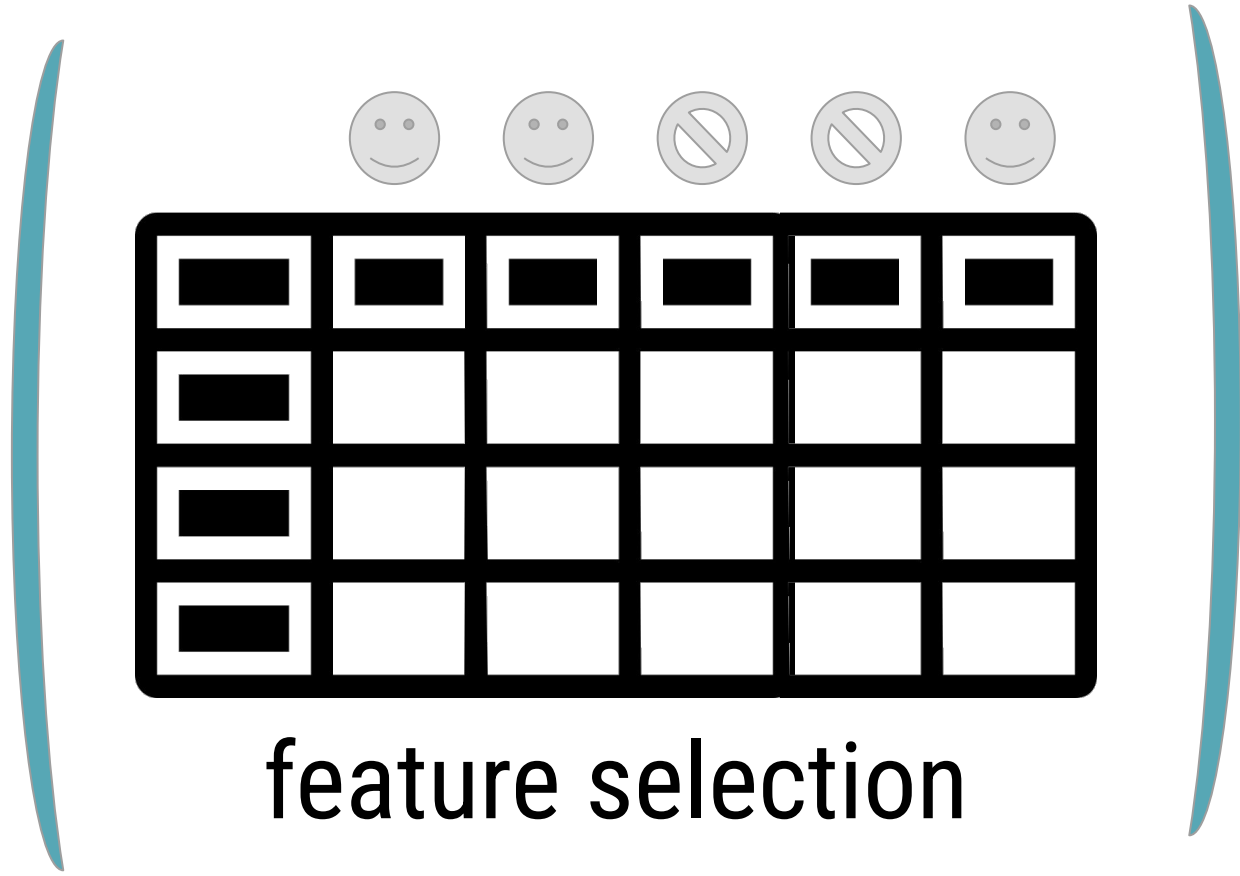


Data partitioning:
Finding the best model
through validation

the data used to build
your predictive model

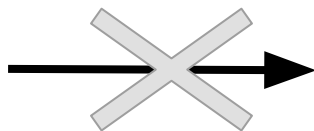
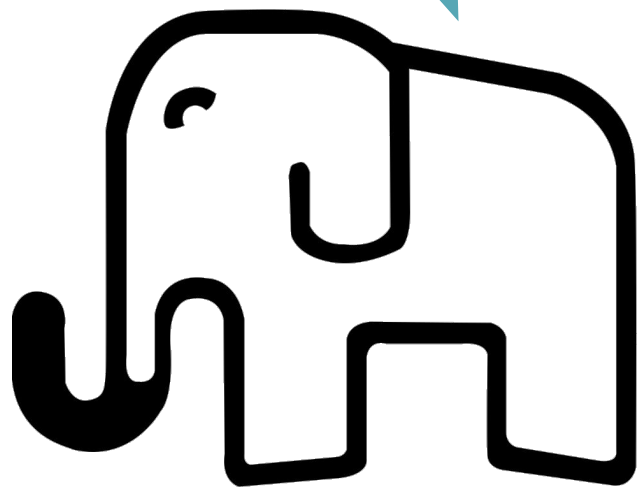
new and independent data set
used to assess if prediction model
is generalizable

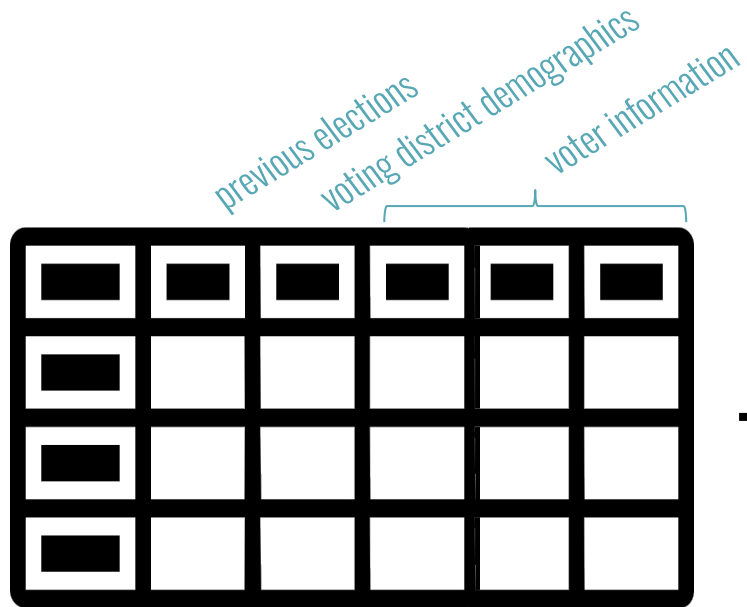




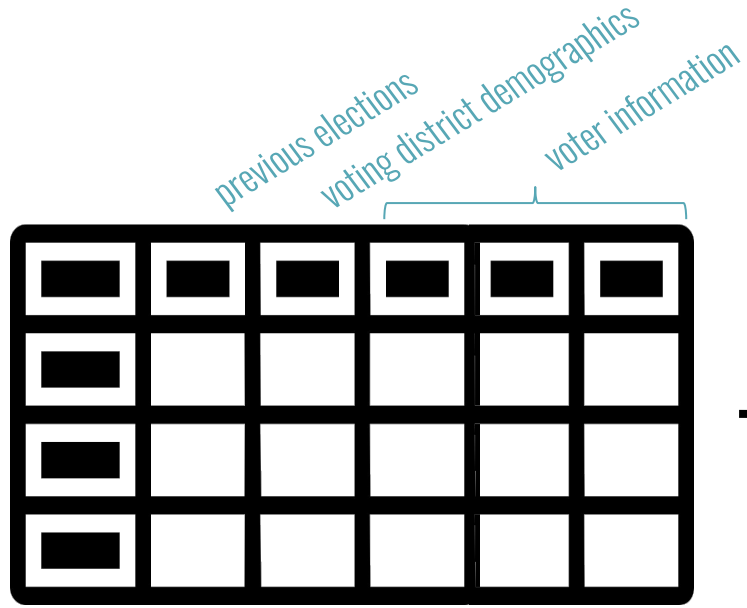
feature selection

elephant height data are likely
not predictive of US elections





these data are likely predictive
of US election outcomes



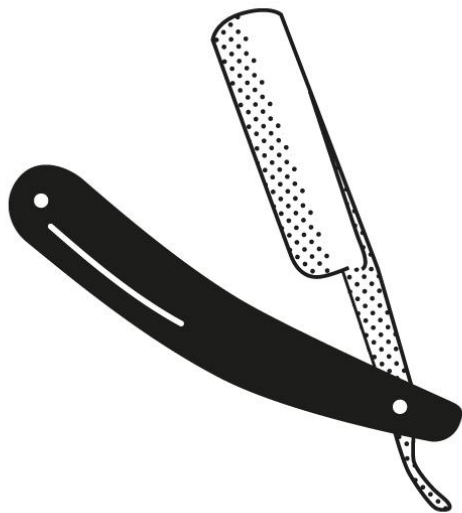
feature selection determines which variables are most predictive and includes them in the model

■	■	■	■	■	■
■					
■					
■					

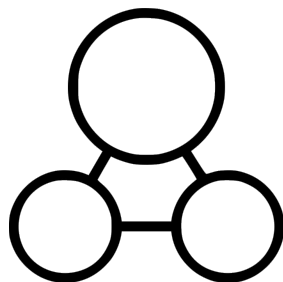


variables that can be used for accurate prediction exploit the relationship between the variables but do NOT mean that one causes the other

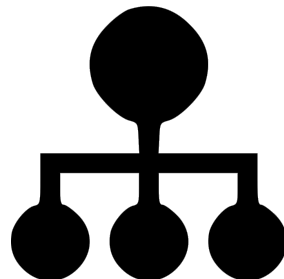
Note: Occam's Razor



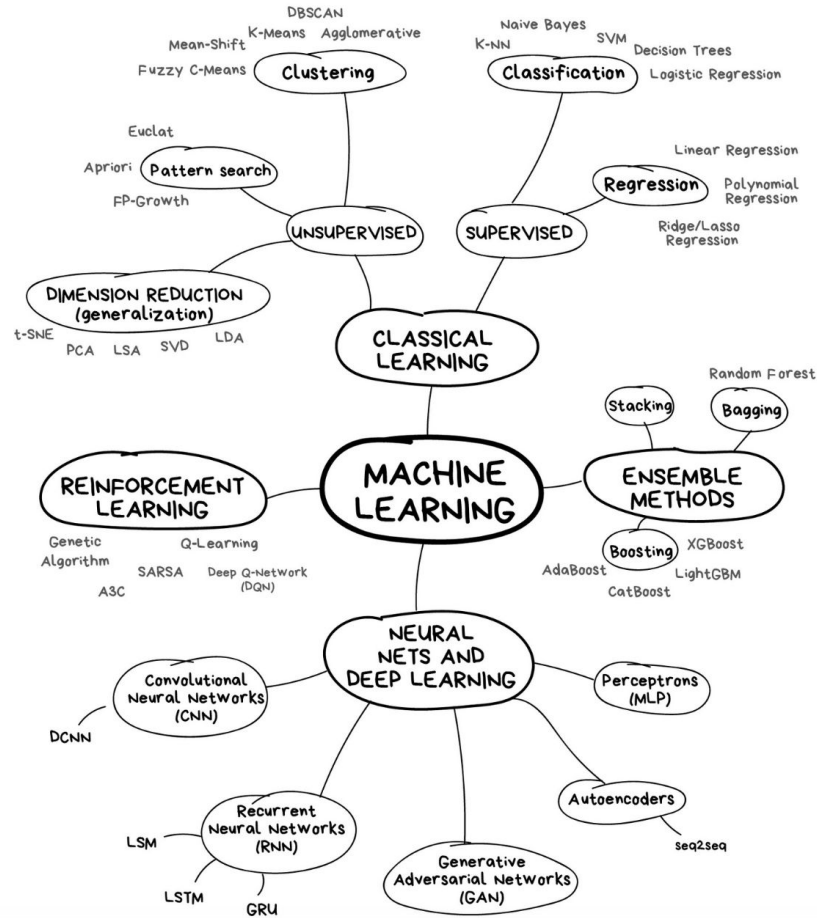
- When faced with two explanations for the *same evidence*, we prefer the explanation that makes the fewest assumptions
- Given *equal performance*, choose the simpler model



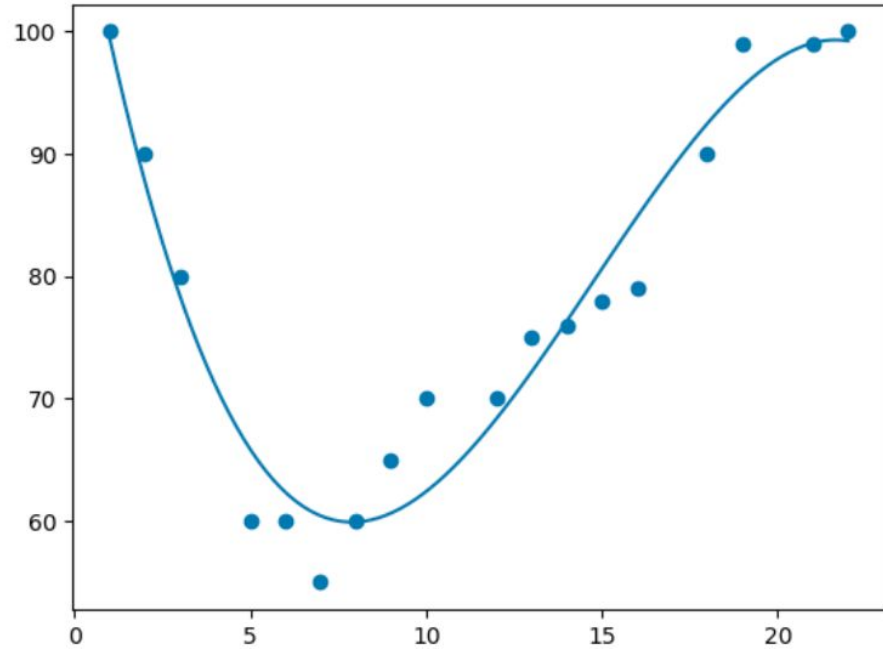
+	-
×	=



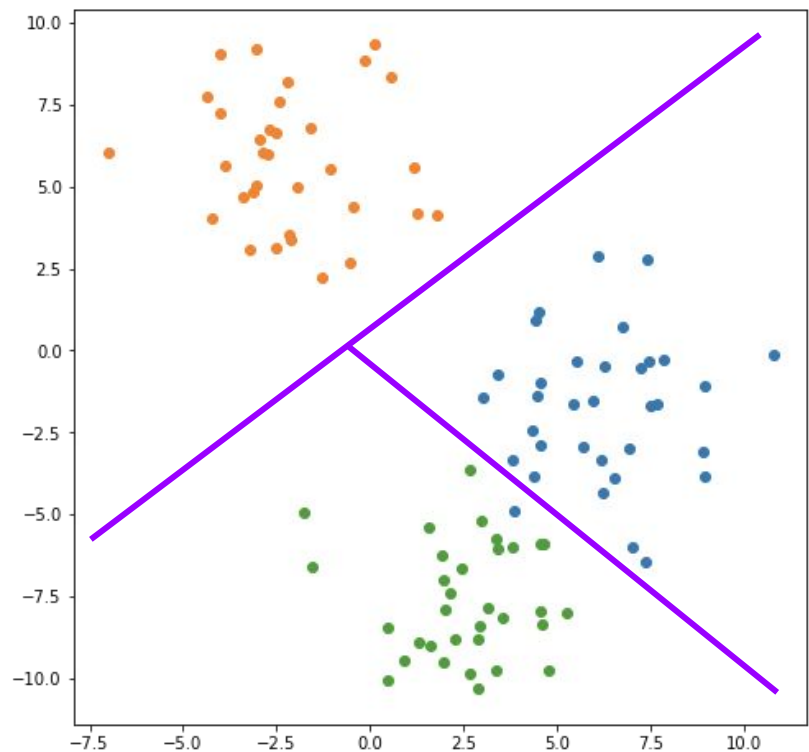
model selection



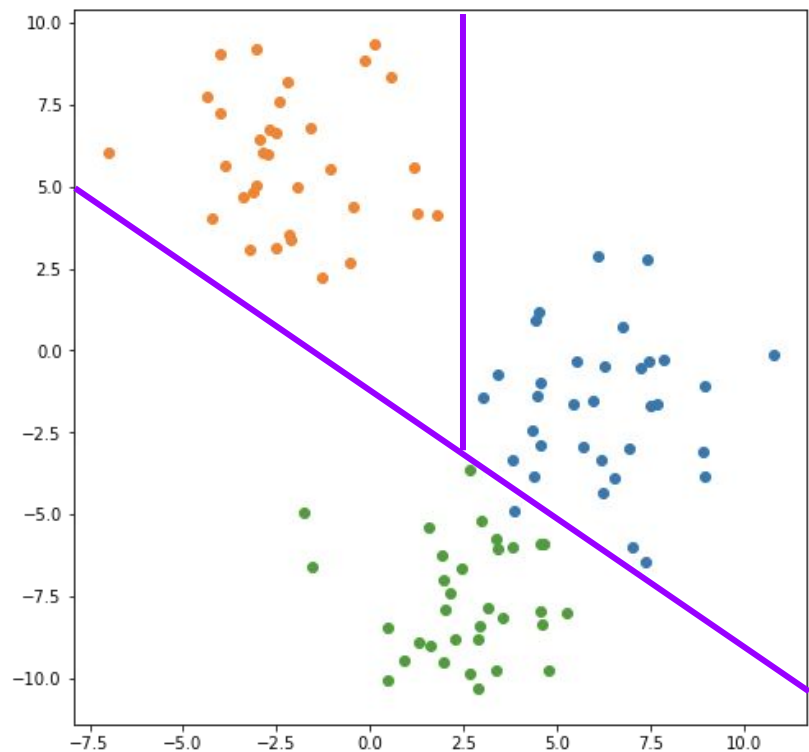
Selecting the “Right” Model



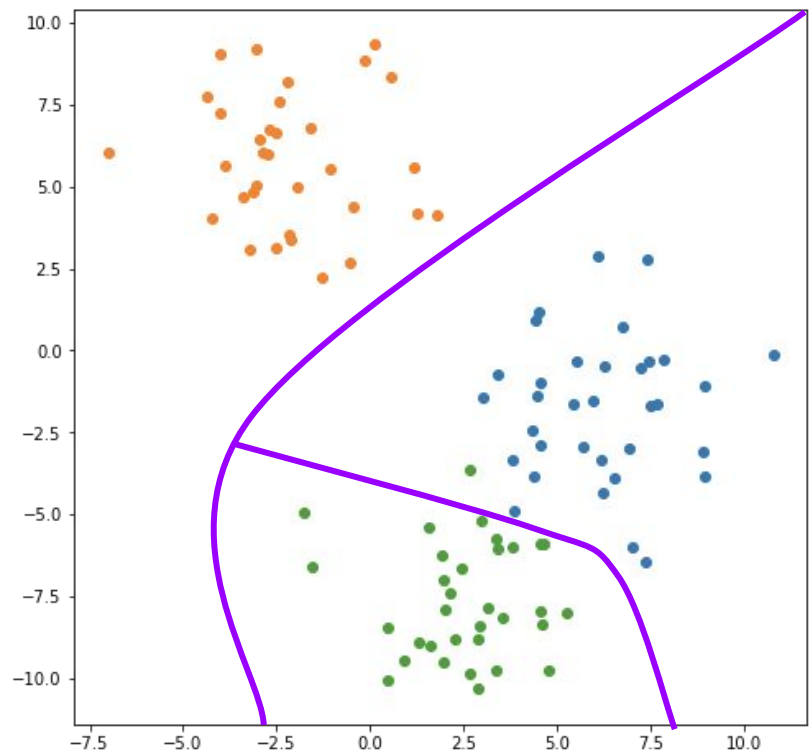
Selecting the “Right” Model



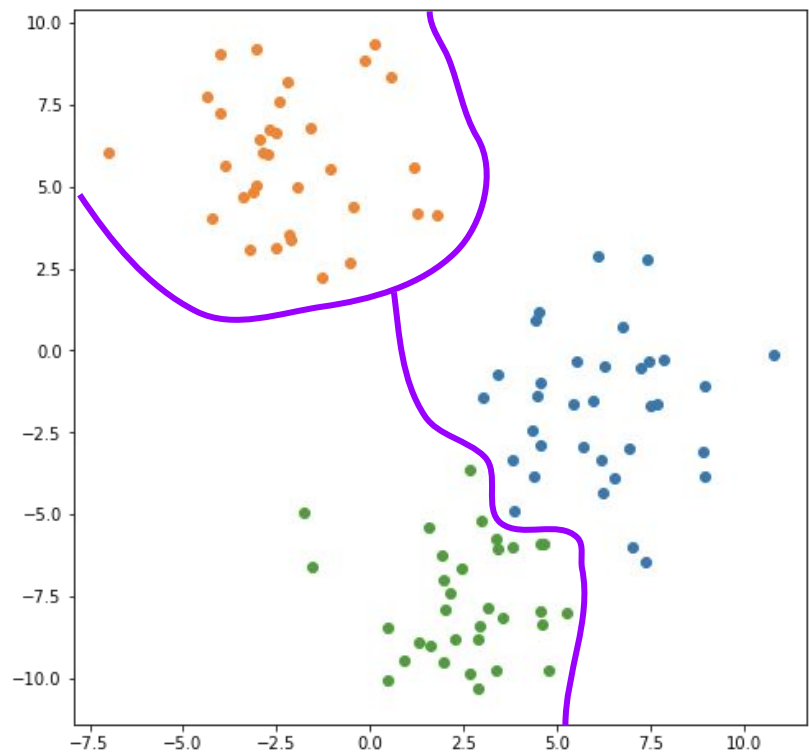
Selecting the “Right” Model



Selecting the “Right” Model



Selecting the “Right” Model



Cross-Validation

In reality, our eyeball meter won't always (and sometimes can't) cut it

Real data are messy and live in dimensions we cannot even comprehend (let alone visualize)

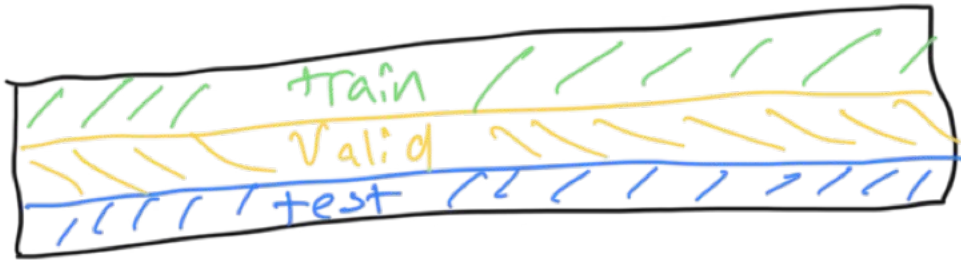
Cross-validation offers a systematic way of assessing various models and determine which ones meet our requirements the closest

- Validation Set
- Leave-one-out
- K-Fold

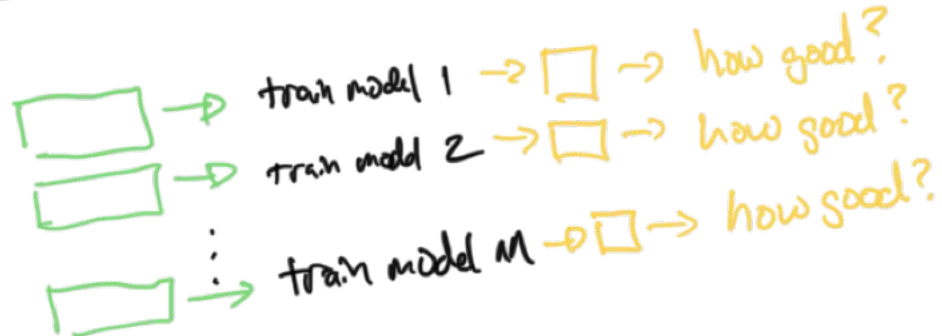
Bigger Datasets. Simpler Models.

① All the data randomly shuffle rows of dataset

② Split into training, validation and test sets



③ if you have M models to try do this once for each



④ Pick best model from validation set performance



train it on $T+V$ then test it on Test set

Model, feature, and hyperparameter selection

Needs: Don't leak information from training/selection process into the test set

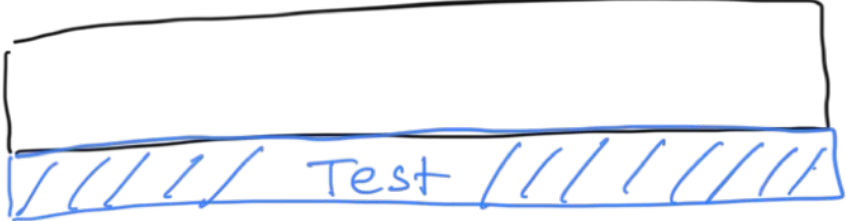
Trade-offs: Usually not enough data to have completely separate train, validation, test sets - where to prioritize?

- Low training data == bad fit
- Low test data == bad selection of model/features/hyperparameters
- Low validation data == poor estimate of generalization performance

Smaller Datasets. Complex Models.

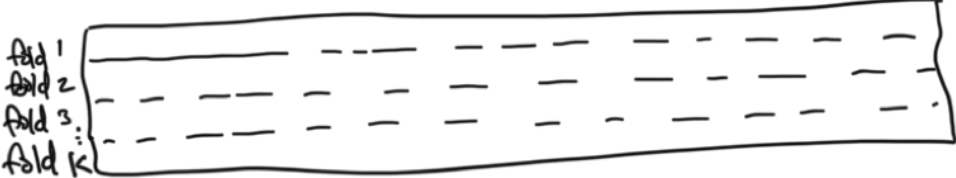
① All the data randomly shuffle rows of dataset

② Split off a test set



A horizontal bar representing a dataset. The bottom portion of the bar is shaded with blue diagonal lines and labeled 'Test' in the center.

③ Split remaining data into K folds with even(ish) amounts of data w/ some(ish) distributions



A horizontal bar representing a dataset, divided into K equal segments by dashed lines. To the left of the bar, the segments are labeled 'Fold 1', 'Fold 2', 'Fold 3', and 'Fold K'.

④ Every split gets its turn as the validation set, use all other splits as training. Validation performance is the mean across splits

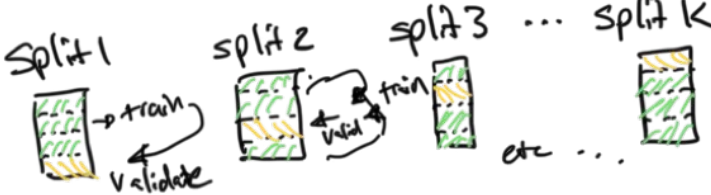
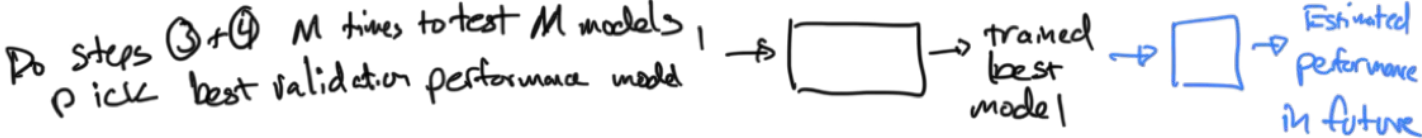


Diagram illustrating the process of training and validation across K splits. It shows four vertical bars representing splits: 'Split 1', 'Split 2', 'Split 3', and 'Split K'. Each bar is divided into two sections: a green section (training) and a yellow section (validation). For 'Split 1', an arrow labeled 'train' points from the green section to the yellow section, and an arrow labeled 'validate' points from the yellow section back to the green section. For 'Split 2', an arrow labeled 'train' points from the green section to the yellow section, and an arrow labeled 'validate' points from the yellow section back to the green section. For 'Split 3' and 'Split K', only the green section is shown. The text 'etc ...' is written between 'Split 3' and 'Split K'.

⑤ Do steps ③ + ④ M times to test M models, pick best validation performance model → trained best model → Estimated performance in future



A flowchart showing the final steps of the process. It starts with a box labeled 'trained best model', which points to a box labeled 'Estimated performance in future'.



model assessment

Many Metrics!

Assessing machine learning models involves gauging its performance on novel data

- Novel data can come from testing or validation sets
 - Validation → Fine-tuning parameters
 - Test → Final assessment before deployment
- Performance assessments depend on the specific problem/question
- Different metrics measure different aspects of models
 - e.g. MSE/RMSE, MAE, accuracy, specificity/sensitivity, etc.

Accuracy can mislead

If classes are imbalance what would “chance performance” be?

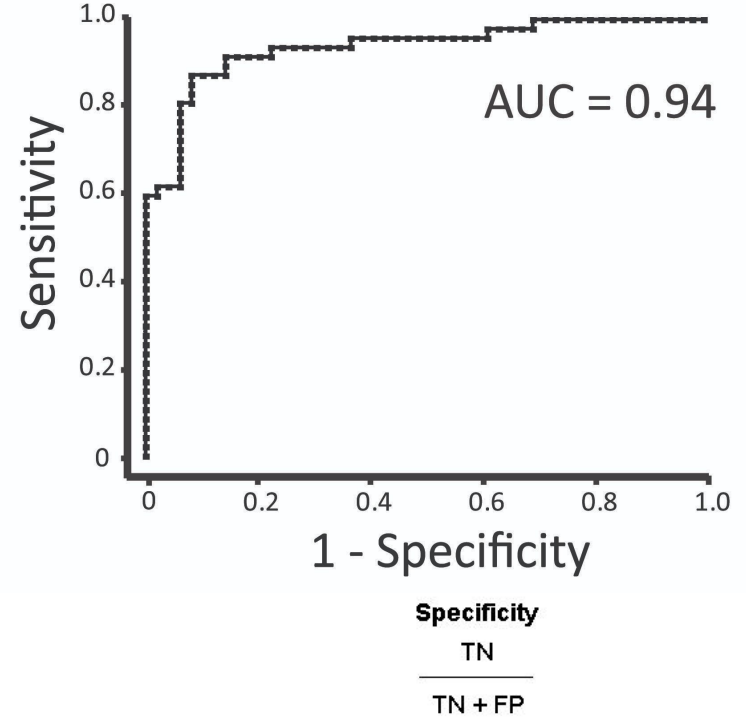
Classic example: Cancer detection

- 1/1000 actually have cancer
- Your prediction algorithm misdiagnoses 1% of people
- Algorithm will tell 10- people who are healthy they are sick (FP) for every 1 person who is actually sick (TP)

		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP)
	Negative	False Negative (FN)	True Negative (TN)

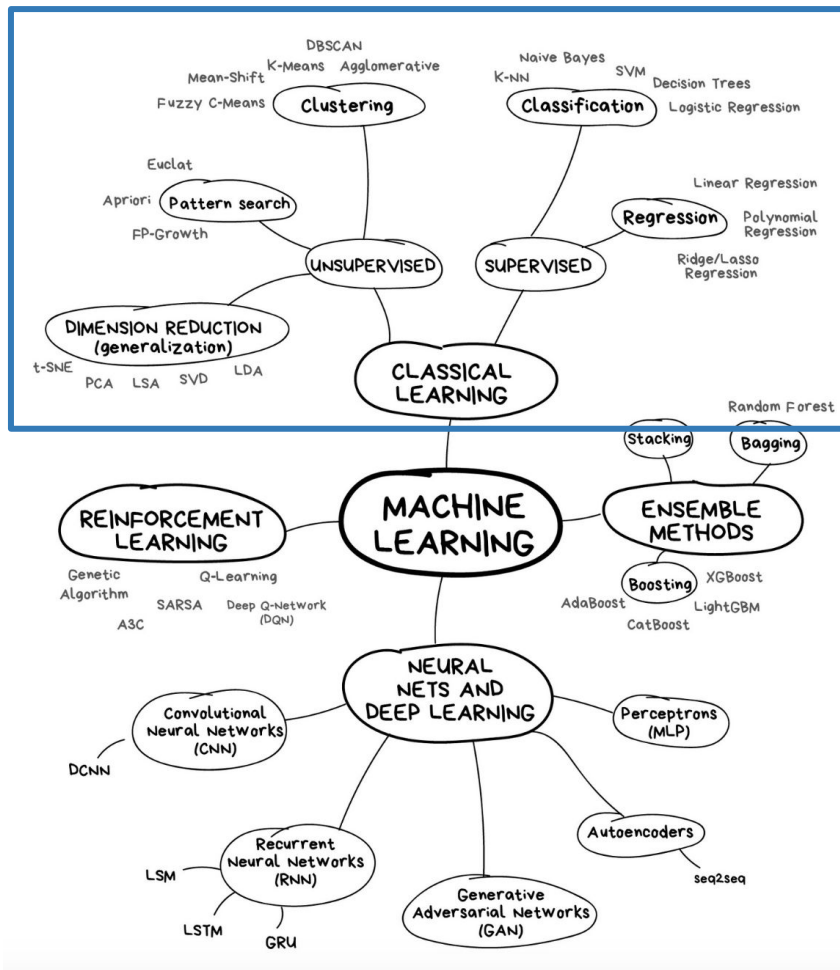
A 2x2 table is a type of confusion matrix

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

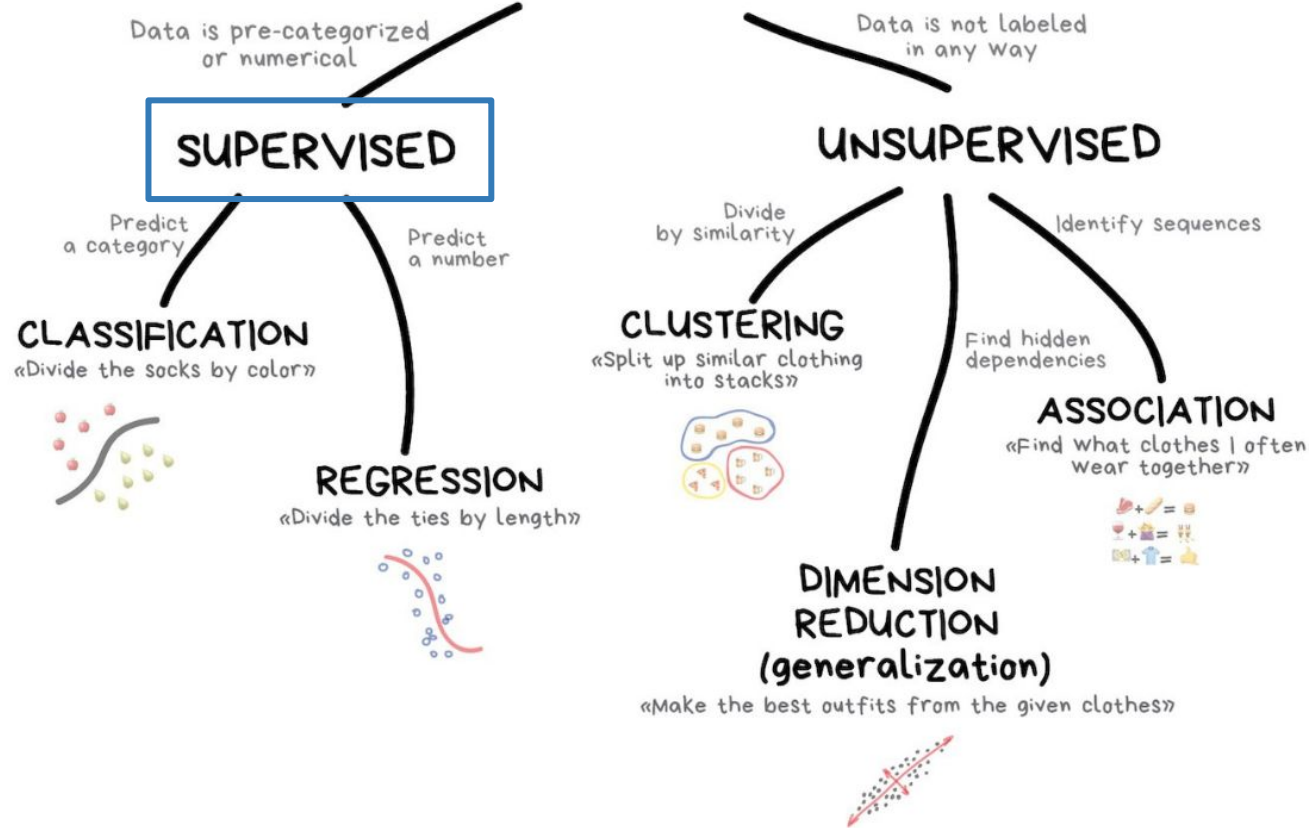


Accuracy	What % were predicted correctly?
Sensitivity	Of those that <i>were</i> positives , what % were predicted to be positive?
Specificity	Of those that were negatives , what % were predicted to be negative?

Classical Machine Learning

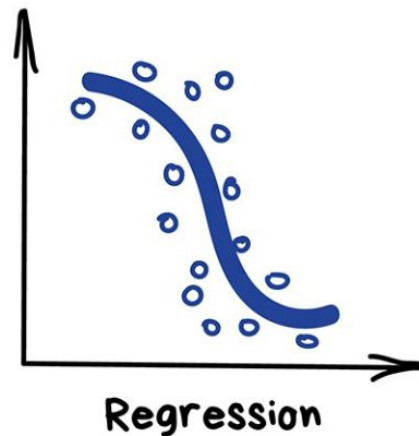


CLASSICAL MACHINE LEARNING

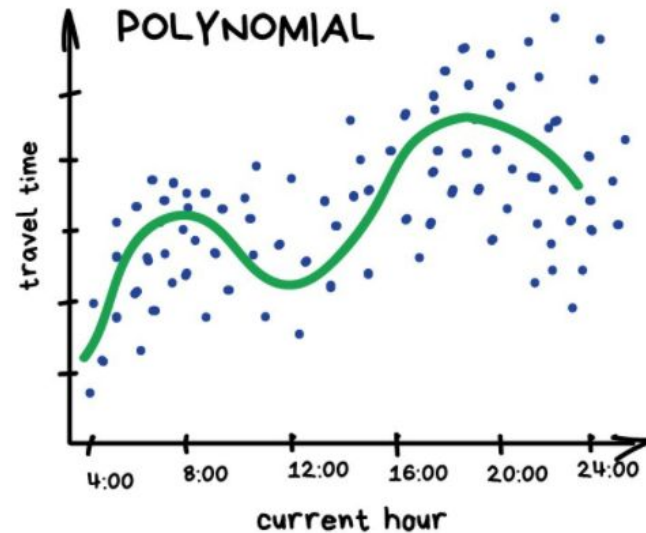
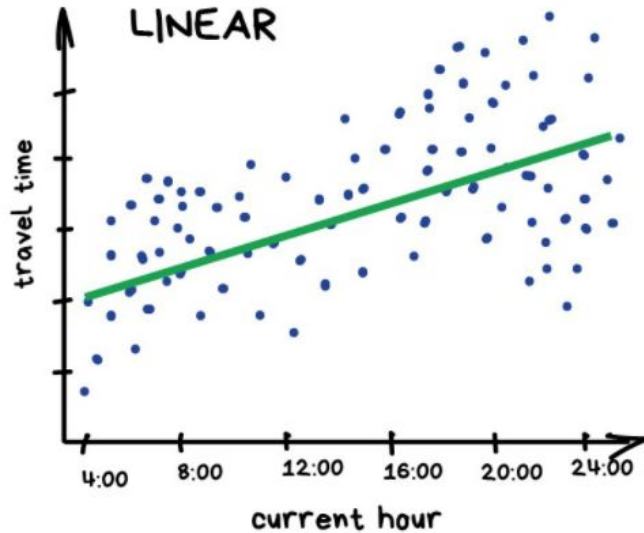


Regression

- Predicting a **continuous** outcome by finding a relationship between inputs and outputs
- Makes predictions on new data based on the learned patterns from the training set
- Used for:
 - Stock price forecasts
 - Demand and sales volume analysis
 - Number-time correlations
- Popular regression models include:
 - Linear regression
 - Polynomial regression



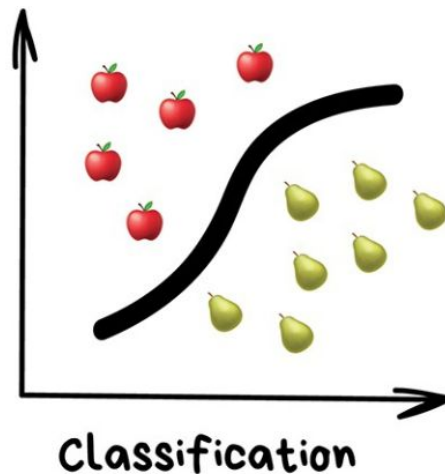
PREDICT TRAFFIC JAMS



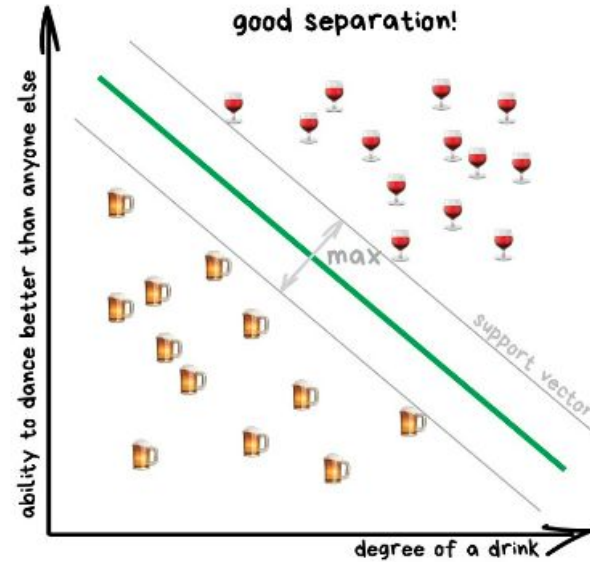
REGRESSION

Classification

- Predicting a **categorical** outcome by finding boundaries between classes of data points
- Makes predictions on new data based on where within the decision boundary it falls
- Used for:
 - Spam filtering
 - Fraud detection
- Popular classification models include:
 - Support Vector Machine
 - Decision Tree
 - K-Nearest Neighbors

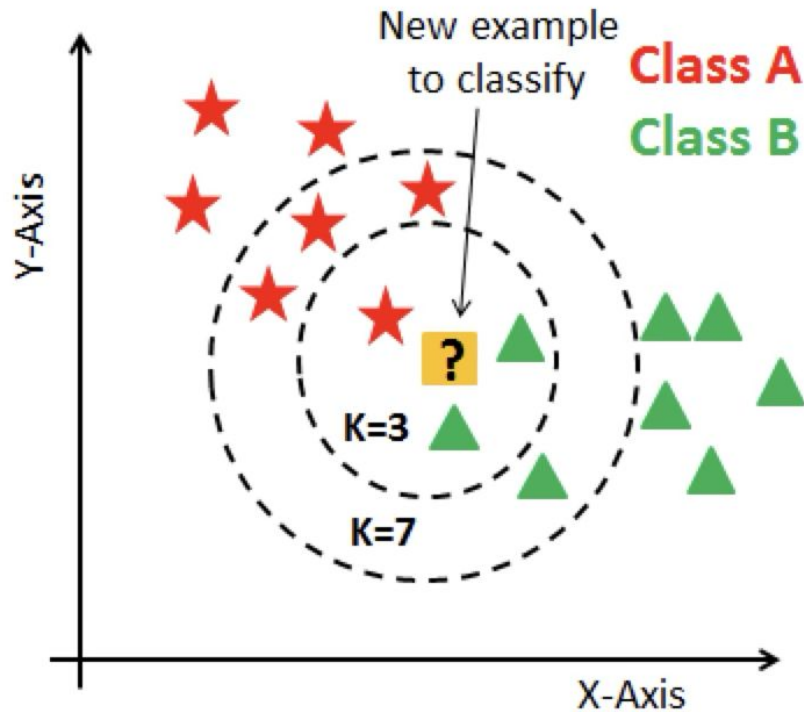


SEPARATE TYPES OF ALCOHOL

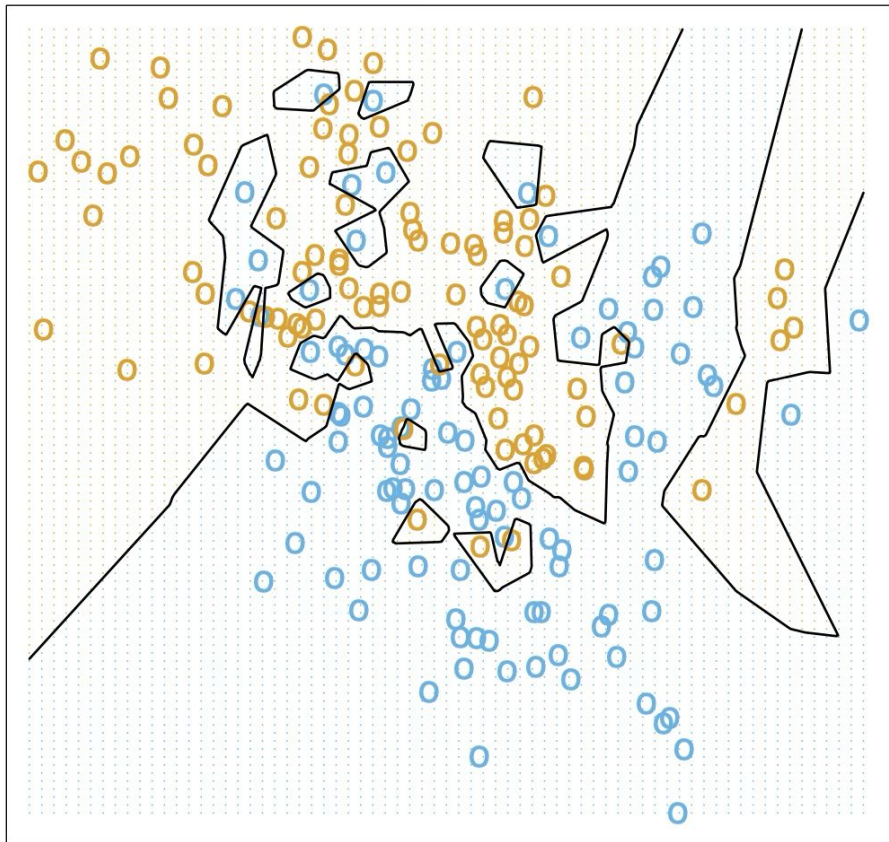


SUPPORT VECTOR MACHINE

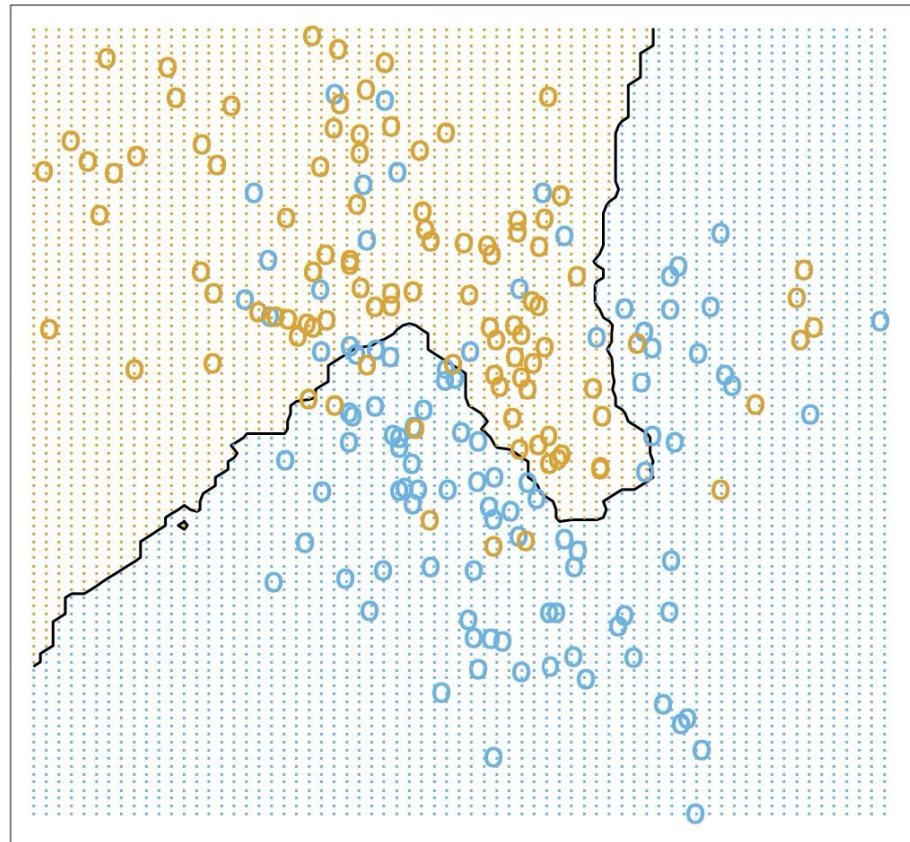
K-nearest neighbors



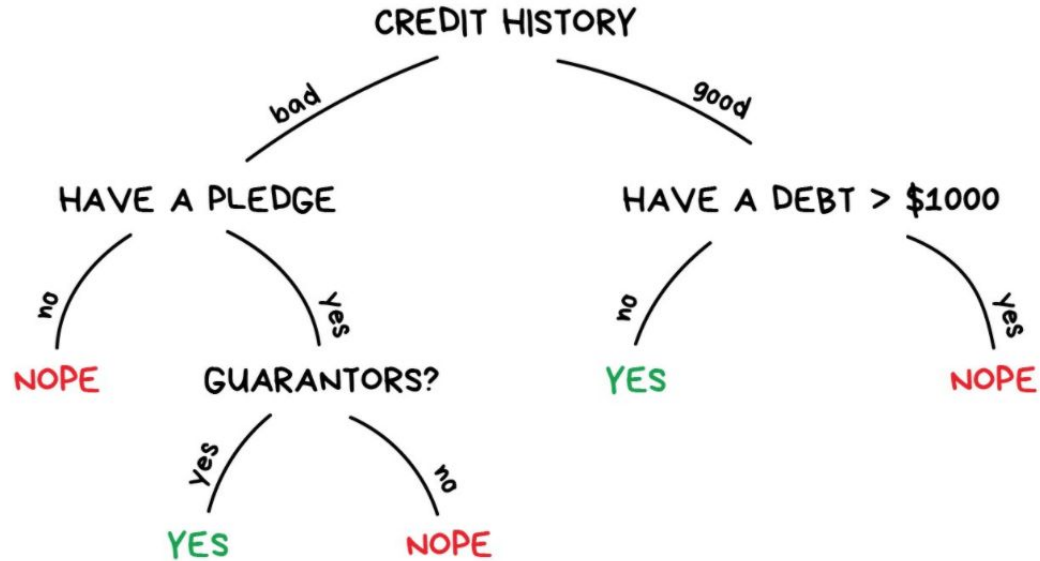
1-Nearest Neighbor Classifier



15-Nearest Neighbor Classifier



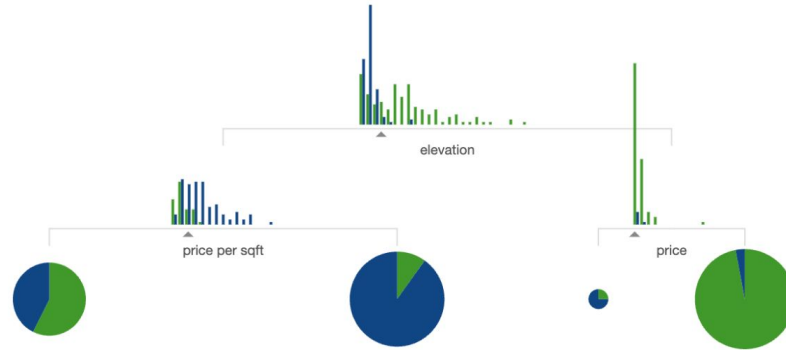
GIVE A LOAN?



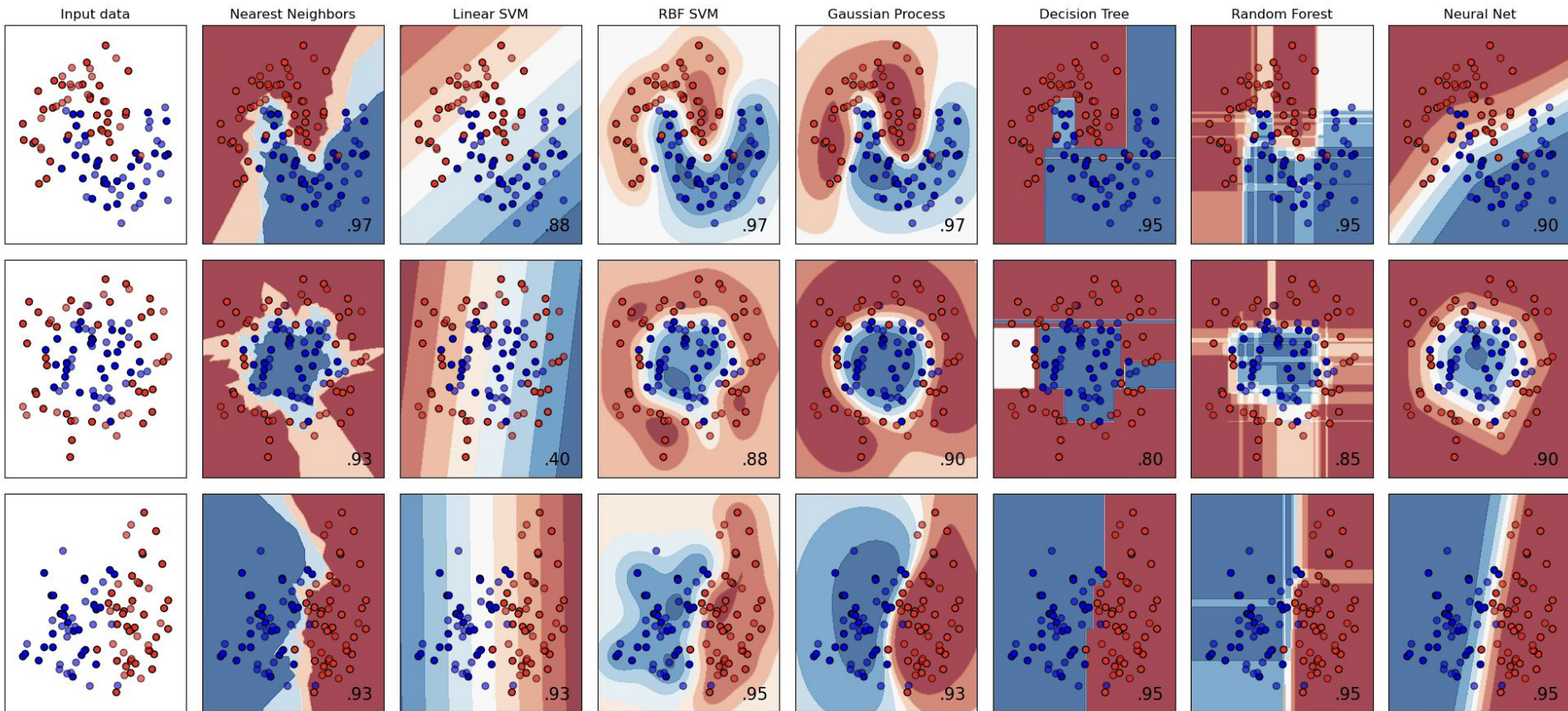
DECISION TREE

Growing a tree

Additional forks will add new information that can increase a tree's **prediction accuracy**.



<http://www.r2d3.us/visual-intro-to-machine-learning-part-1/>



Preventing overfitting is important!

When AI models make errors it is typically because the dataset is a bad sample, leading to increased likelihood of overfitting patterns that are particular to this bad sample.

Generically there are two issues at play here.

Bad procedures (or carelessness) led to a bad model

Then ethical failings led to this bad model being put into service and kept in service.



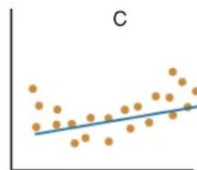
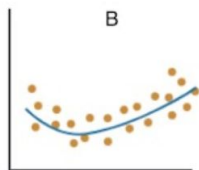
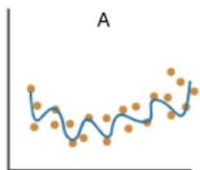
Model Fit

<https://forms.gle/eE9XXZoUbflowLWYp6>

Which regression model is (in your opinion) over fit, under fit, or best fit?

0 points

Regression



Under fit

Best fit

Over fit

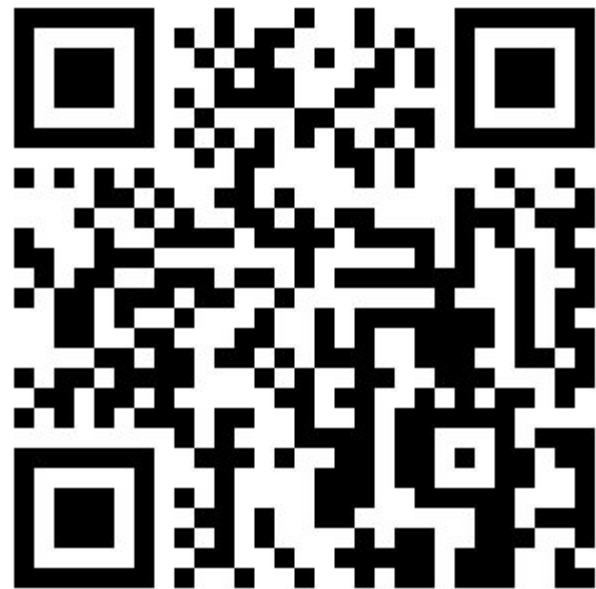
Model A - 9th order
polynomial

☐☐☐

Model B - 3rd order
polynomial

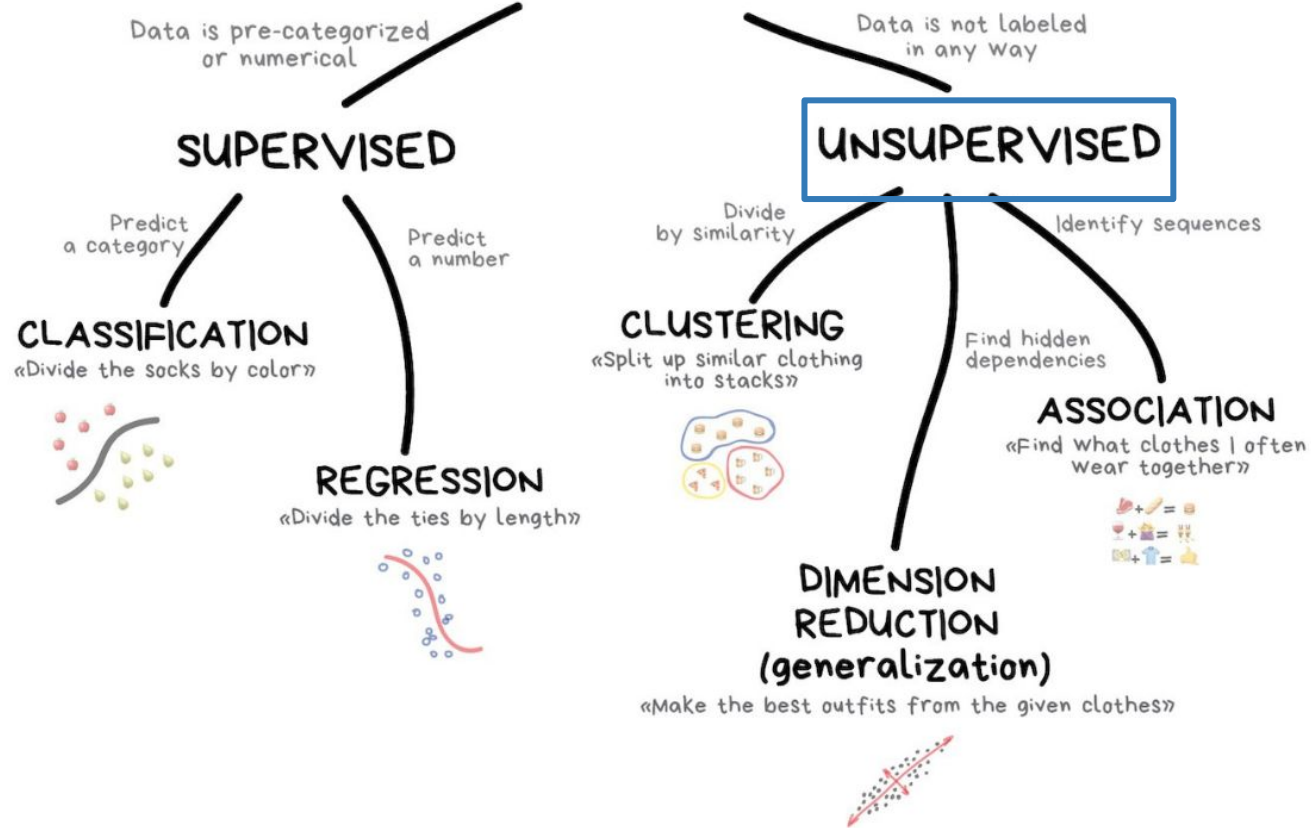
☐☐☐

Model C - 1st order
polynomial

☐☐☐

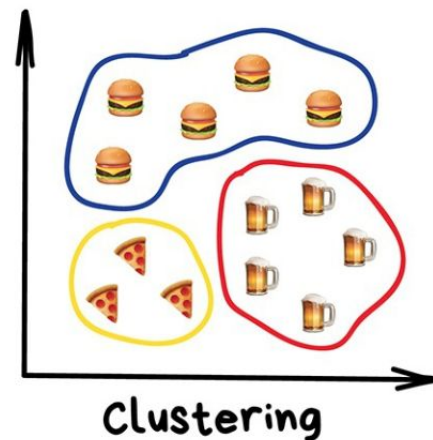
Unsupervised Machine Learning

CLASSICAL MACHINE LEARNING



Clustering

- Groups data points into clusters based on how **similar** they are to one another
- Similarity is usually defined by spatial relationships
- Used for:
 - Analyze and label new data
 - Anomaly detection
 - Merging points on a map
- Popular clustering models include:
 - K-means clustering
 - Gaussian Mixture
 - DBSCAN

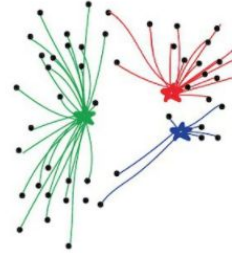


PUT KEBAB KIOSKS IN THE OPTIMAL WAY

(also illustrating the K-means method)



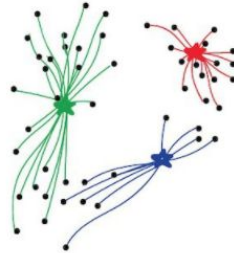
1. Put kebab kiosks in random places in city



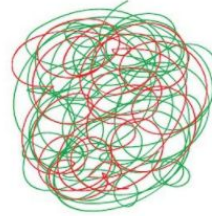
2. Watch how buyers choose the nearest one



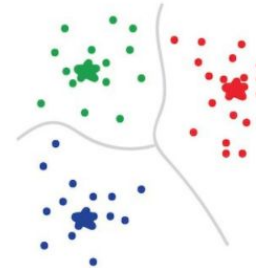
3. Move kiosks closer to the centers of their popularity



4. Watch and move again



5. Repeat a million times

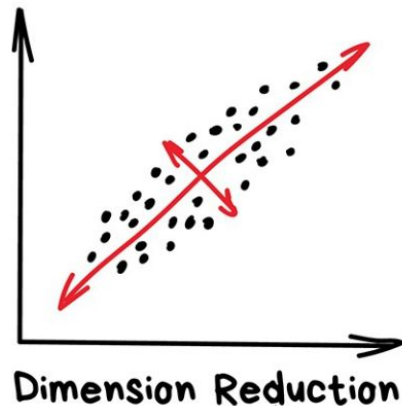


6. Done!
You're god of kebabs!

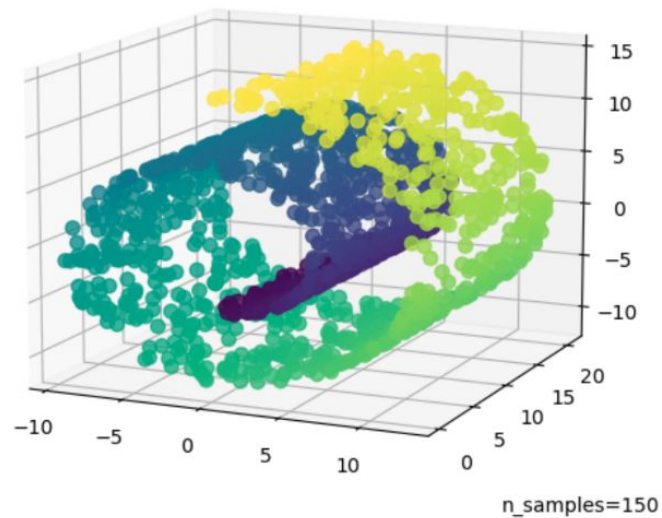
K-Means

Dimensionality Reduction

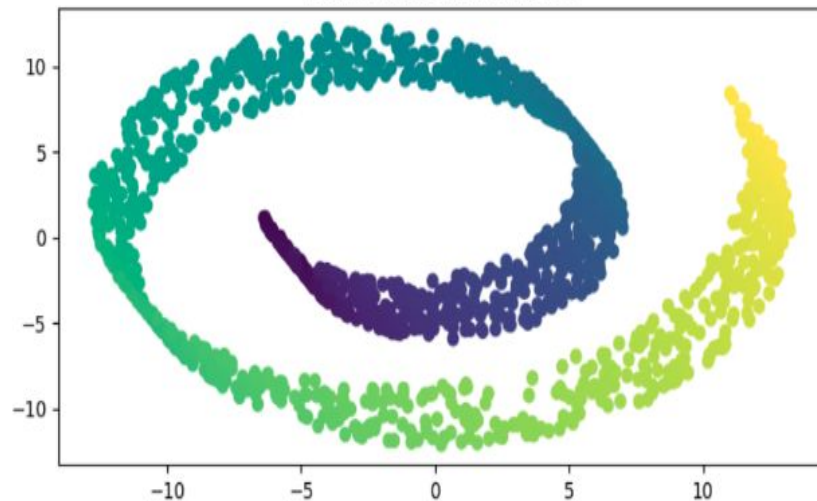
- Reduce high-dimensional data down to fewer dimensions
- Transforms data in a manner that preserves important, explanatory information
- Used for:
 - Fake image analysis
 - Data compression/feature selection
 - Genomics and bioinformatics
- Popular dimensionality reduction models include:
 - Principal component analysis (PCA)
 - t-SNE
 - UMAP



Swiss roll dataset



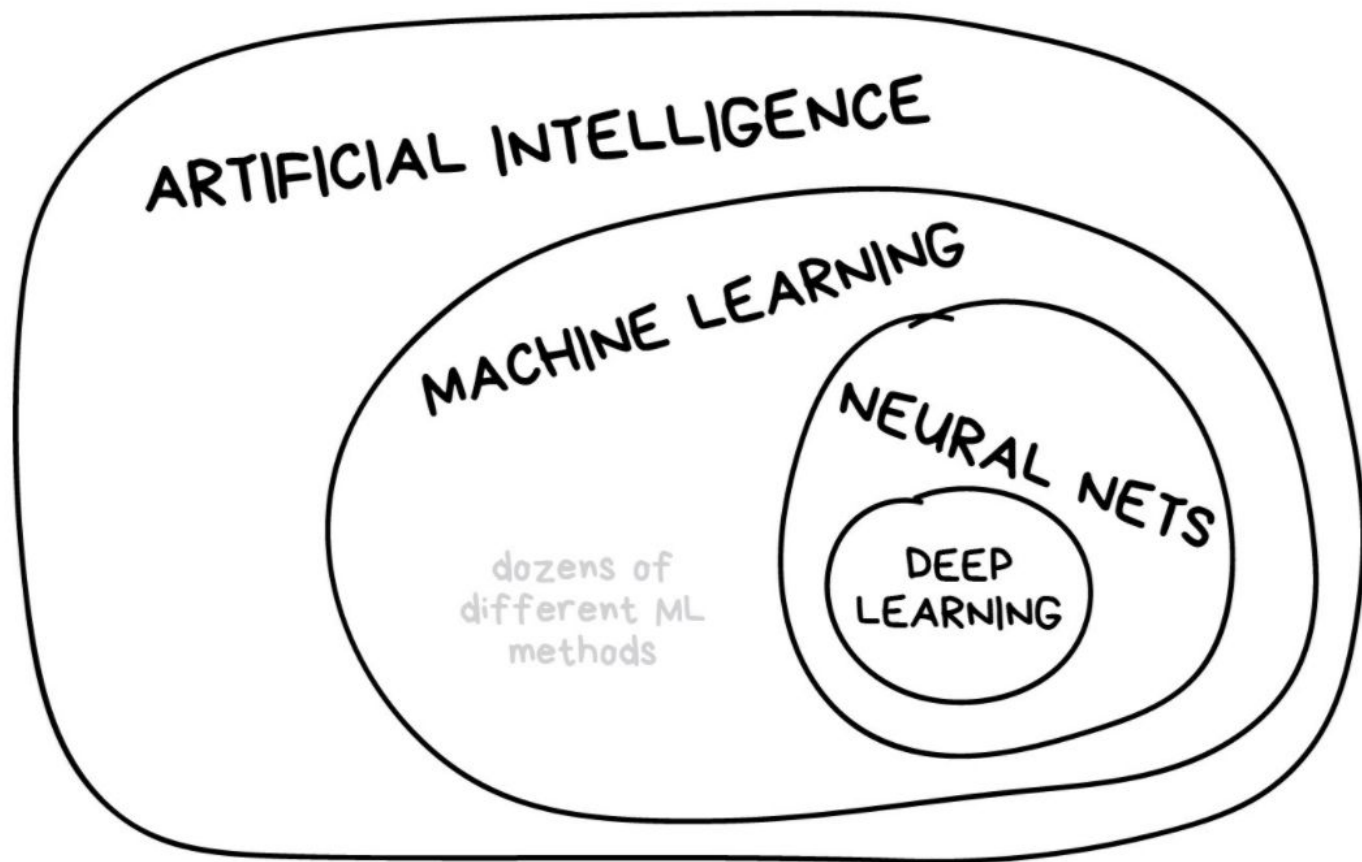
PCA of Swiss Roll Dataset

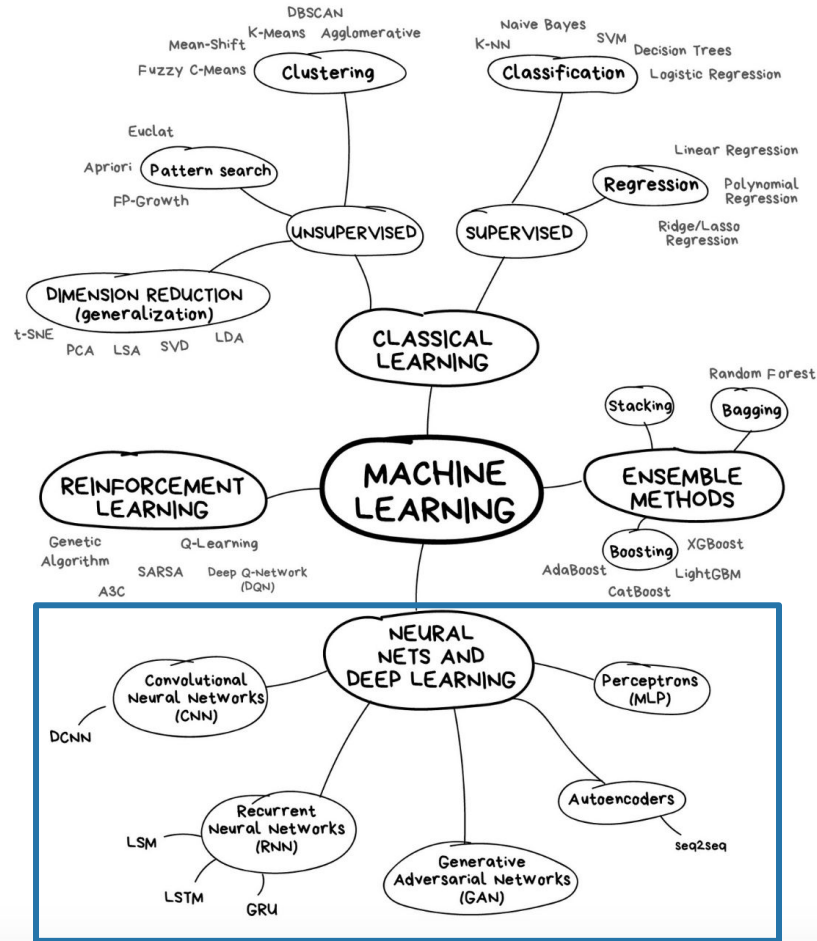
*PCA*

Putting it all together: Prediction Approach



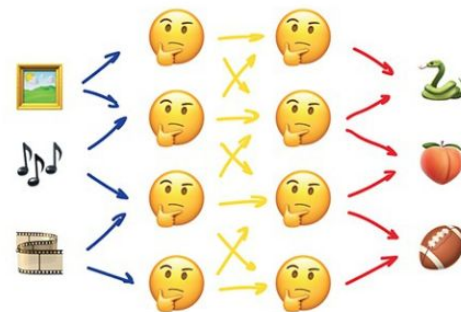
Deep Learning and Neural Networks



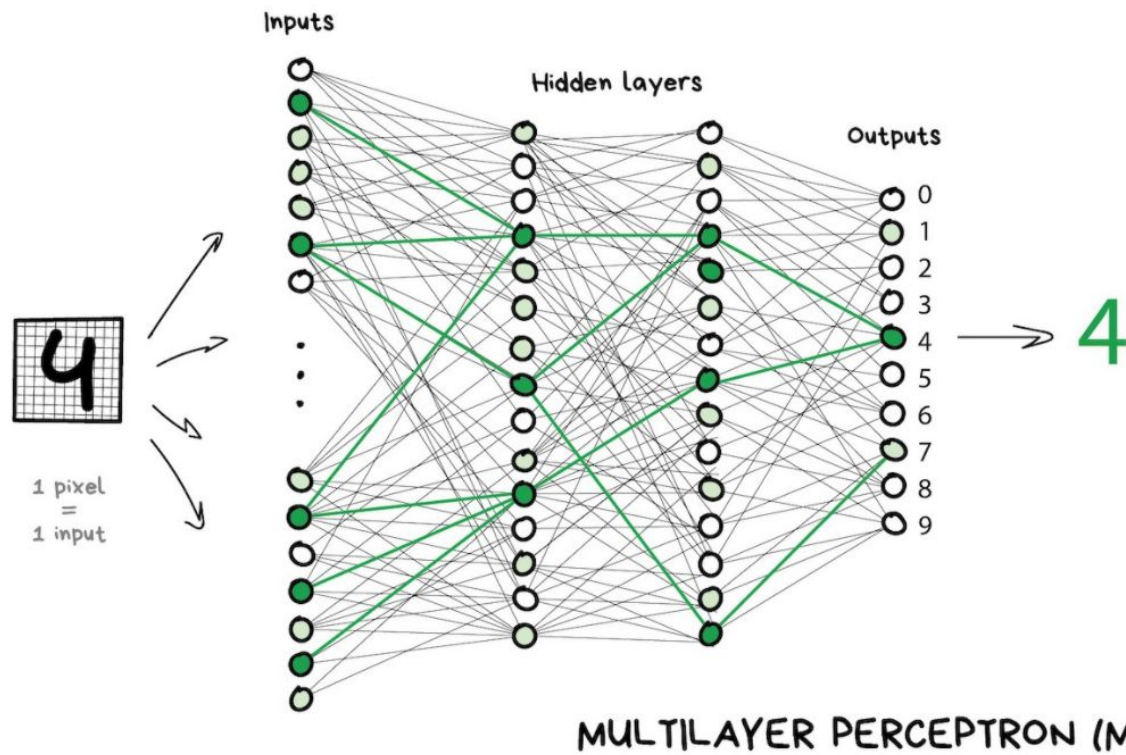


Neural Networks

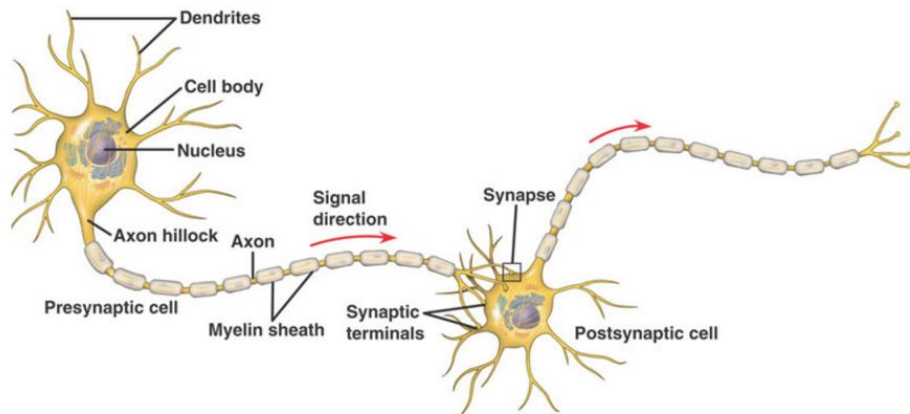
- An arrangement of interconnected nodes inspired by the human brain that process information to learn and make predictions
- **Can** replace all the algorithms we've discussed
- Used for:
 - Image segmentation
 - Speech recognition
 - Image/text/video generation
- Popular neural network architectures include:
 - Multi-layer Perceptron
 - Convolutional Neural Networks
 - ResNet



Neural Networks

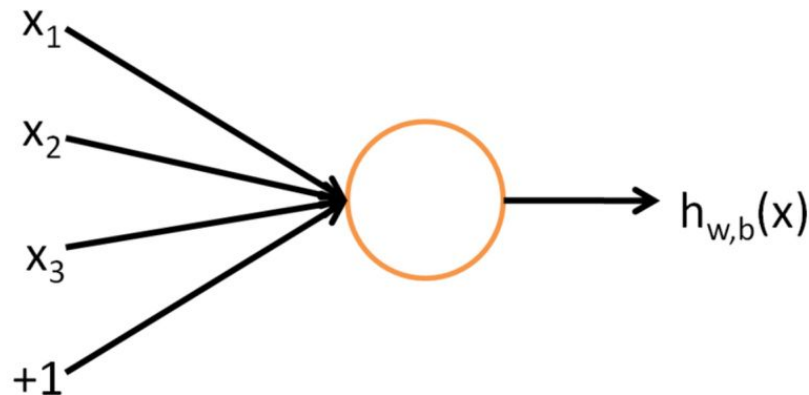


WHAT IS A NEURON?



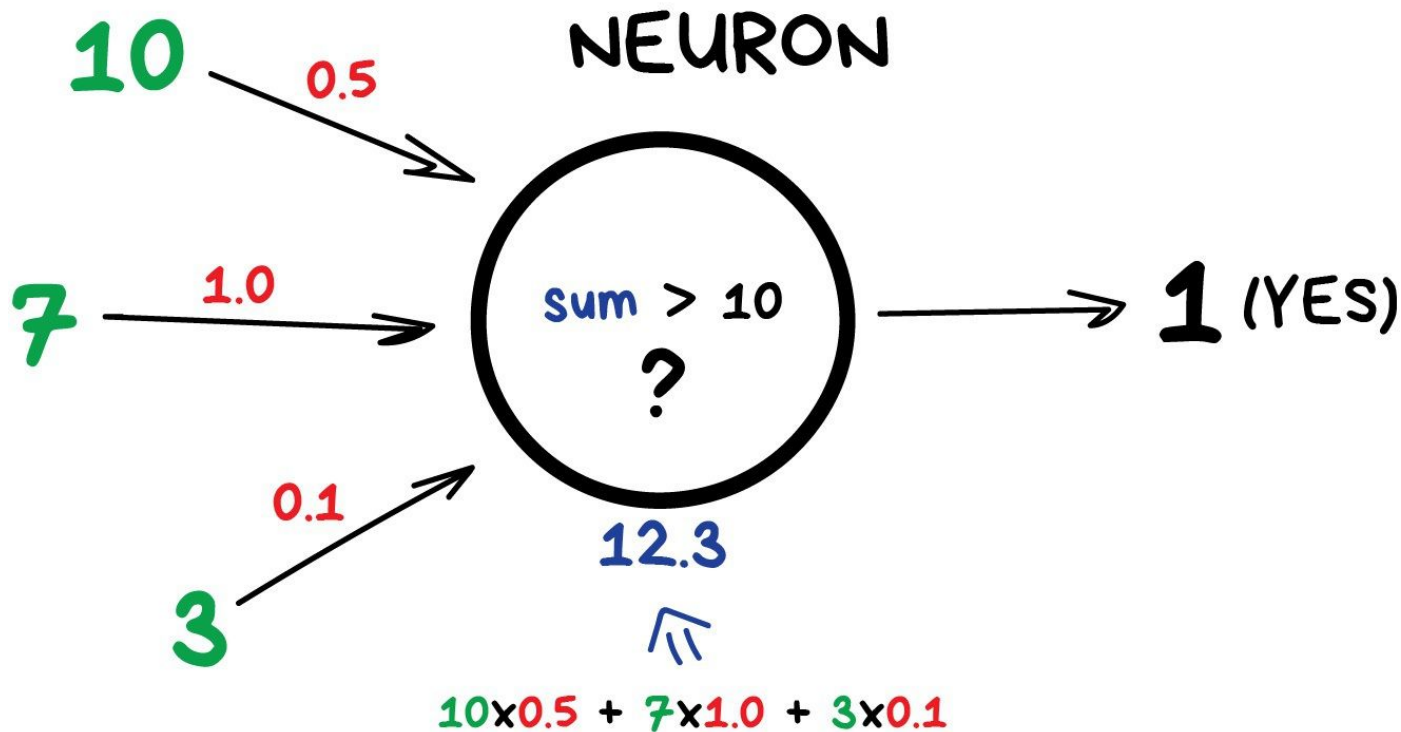
- Receives signal on synapse
- When trigger sends signal on axon

MATHEMATICAL NEURON



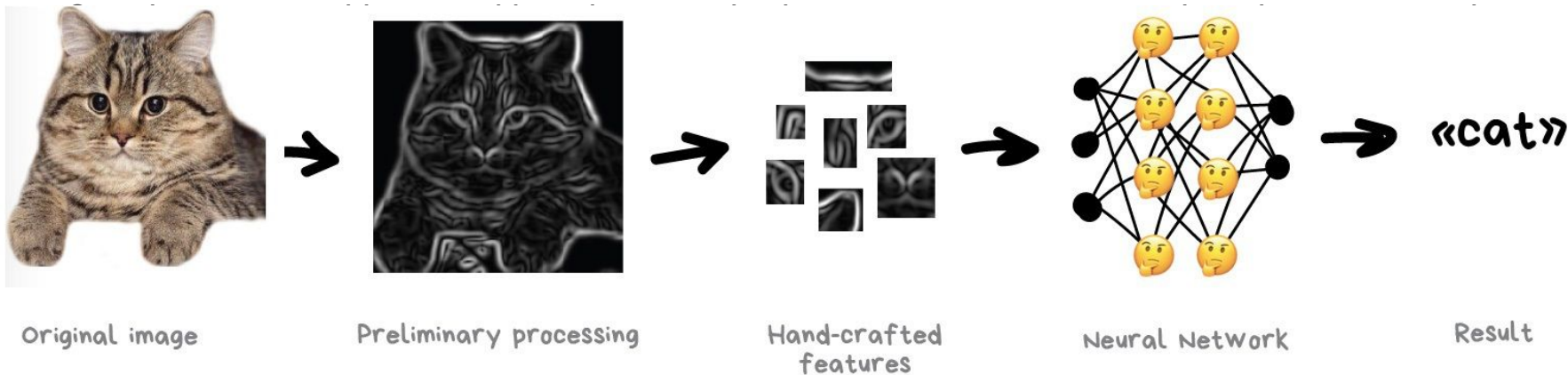
- Mathematical abstraction, inspired by biological neuron
- Either on or off based on sum of input

These weights tell the neuron to respond more to one input and less to another. Weights are adjusted when training — that's how the network learns. Basically, that's all there is to it.

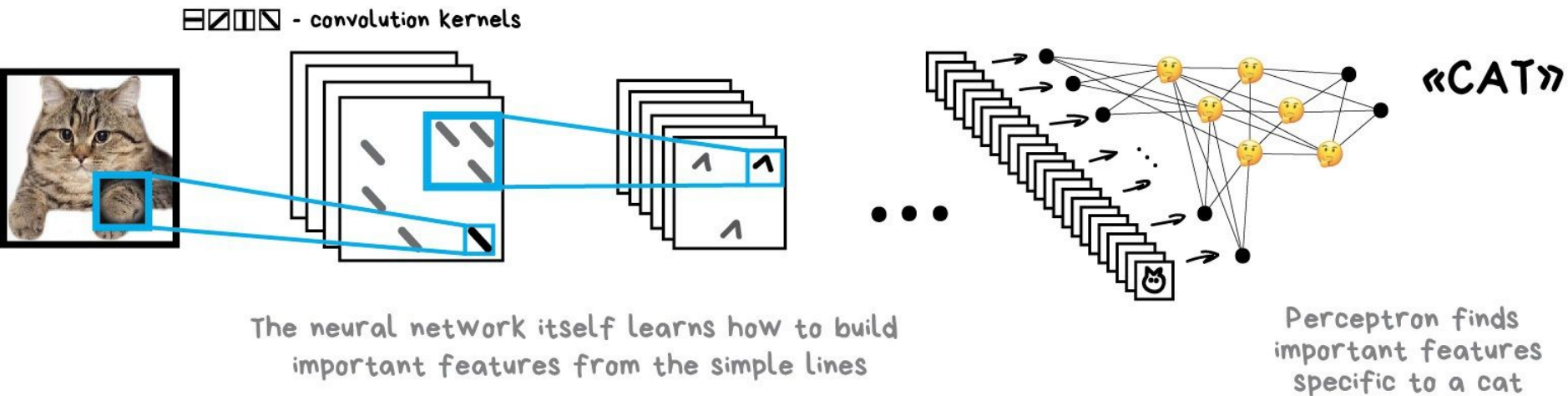


The challenges of image processing

- Extracting features from images is **hard**
 - Historically, humans would manually label images with particular features → handcrafting features



CNNs to the rescue!



“CNNs are all the rage right now. They are used to search for objects on photos and in videos, face recognition, style transfer, generating and enhancing images, creating effects like slow-mo and improving image quality. Nowadays CNNs are used in all the cases that involve pictures and videos.”

CONVOLUTIONAL NEURAL NETWORK (CNN)

HOW A DEEP NEURAL NETWORK SEES

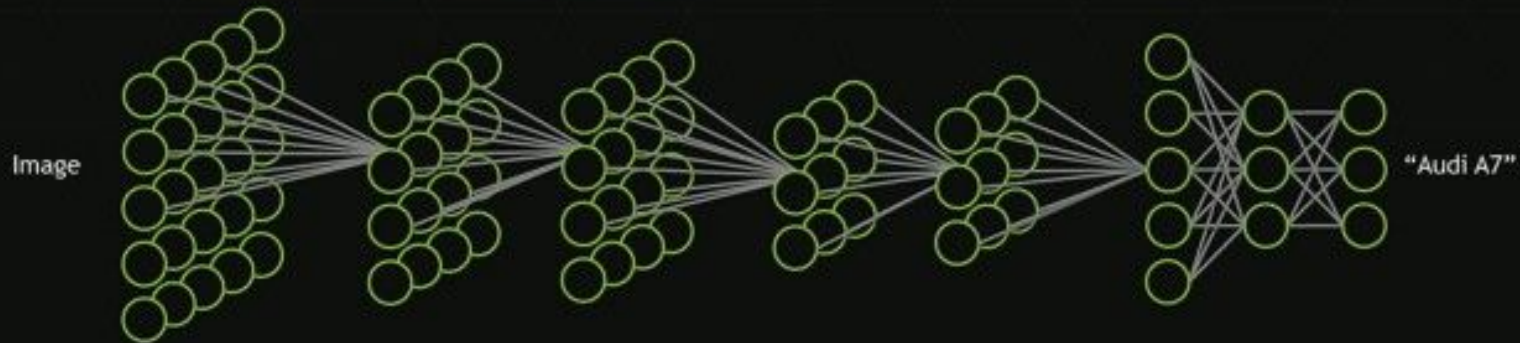


Image source: "Unsupervised Learning of Hierarchical Representations with Convolutional Deep Belief Networks" ICML 2009 & Comm. ACM 2011, Honglak Lee, Roger Grosse, Rajesh Ranganath, and Andrew Ng.

Machine Learning Ethics

What to do about ethical concerns and biased algorithms...

1. Anticipate and plan for potential biases before model generation. Check for bias after.
2. Have diverse teams.
3. Use machine learning to improve lives rather than for punitive purposes.
4. Revisit your models. Update your algorithms.
5. You are responsible for the models you put out into the world, unintended consequences and all.

Think about whether the models you're building should even be built.

Predictive algorithms should (*at a minimum*) be FAT

Fair: lacking biases which create unfair and discriminatory outcomes

- For whom does this algorithm fail?
- Steps to take:
 1. Verify data about individual is correct
 2. Carry out “sensitivity test”

Accountable/Accurate: answerable to the people subject to them

- Correct data used? Is there a mechanism for appeal?

Transparent: open about how and why particular decisions were made

- Think *carefully* about what transparency is (Handing over source code likely isn't the answer)

A Mulching Proposal

Analysing and Improving an Algorithmic System for Turning the Elderly into High-Nutrient Slurry

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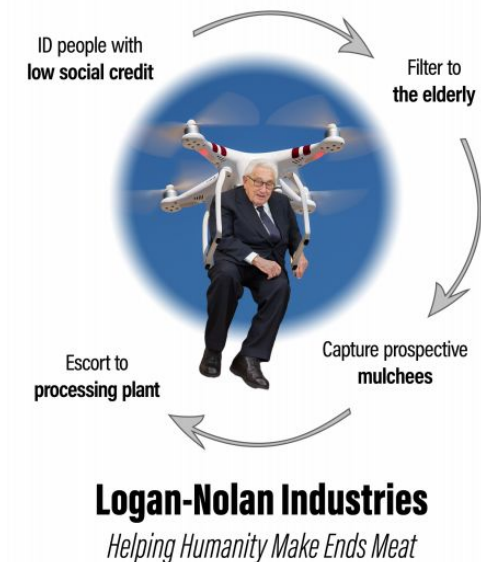
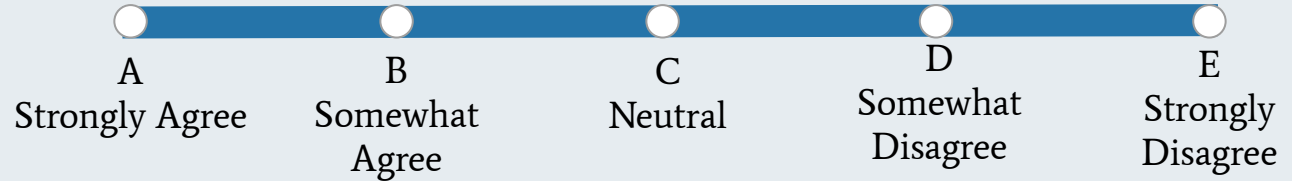


Figure 1: A publicity image for the project, produced by Logan-Nolan Industries

Prediction Thoughts



We should start using this algorithm to mulch up the elderly



A Mulching Proposal

FAIR - equally considers all elderly individuals

ACCURATE - pre- has mechanism for appeal; post - compensation

TRANSPARENT - website with all features; testable

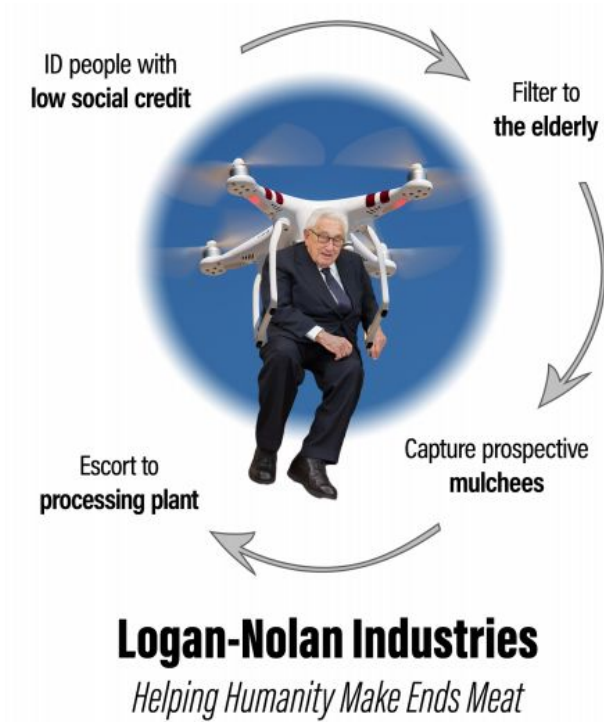
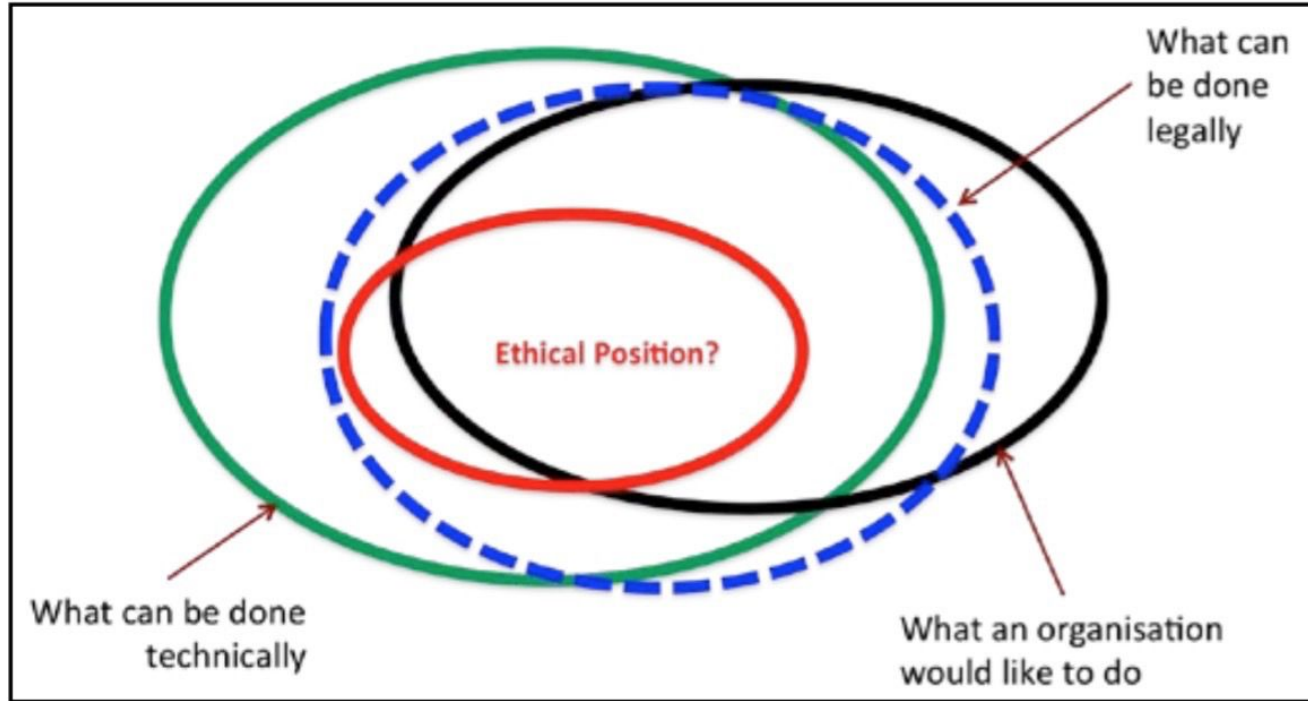


Figure 1: A publicity image for the project, produced by Logan-Nolan Industries

Checklists are helpful, but they're not an excuse for thoughtlessness.

Big Data Ethics



Beyond COGS 108

- **Supervised Learning - COGS 118A**
 - Labelled data, predict, classify, or fit a model
- **Unsupervised Learning - COGS 118B**
 - No labels, no feedback
 - Find hidden structures in a model
- **Reinforcement Learning - COGS 182**
 - No labels or rules; exploring an environment
- **Neural Networks - COGS 181**
 - Deep Learning and CNN architectures
- **Genetic Algorithms - COGS 186**
 - Mimicking evolution in an agent through trial-and-error